

FPT SCHOOL OF BUSINESS
& TECHNOLOGY

MASTER THESIS ON SOFTWARE ENGINEERING

HYBRID RETRIEVAL AND LEGAL-DOMAIN ADAPTATION FOR VIETNAMESE LEGAL INFORMATION RETRIEVAL

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Acknowledgements and Declaration of Authorship

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Declaration of Authorship

I declare that this thesis is my own work and has been prepared in accordance with accepted standards of academic integrity. All sources and materials used in this research have been appropriately acknowledged and cited.

The work presented in this thesis has not been submitted, either in whole or in part, for any other degree or academic qualification at this or any other institution.

I take full responsibility for the content, accuracy, and academic integrity of this thesis.

Nghiem Phan Thanh Tuan – 2026

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List of Abbreviations

Terminology follows the retrieval literature and the task definitions introduced in Chapters 3–4 [1, 2, 3, 4].

Abbreviation	Meaning	Explanation
ALQAC	Automated Legal Question Answering Competition	A benchmark and competition for evaluating legal question answering and information retrieval systems.
BCA / MPS	Bộ Công an / Ministry of Public Security collection	A legal question–document collection derived from public questions and published answers associated with Vietnam's Ministry of Public Security.
BM25	Best Matching 25 lexical ranking function	A term-based retrieval method that ranks documents according to query-term matches, term frequency, and document length.
BGE-M3	BGE-M3 multilingual embedding model	A multilingual embedding model used in this thesis as a dense retrieval component.
CV	Cross-validation	An evaluation procedure that partitions data into subsets for model selection and performance estimation.
E5	E5 text-embedding model family	A family of dense embedding models; this thesis evaluates the `multilingual-e5-base` variant.
FT / PT	Fine-tuned / Pretrained	Distinguishes a task-adapted model from its original pretrained version.
IR	Information Retrieval	The task of retrieving and ranking documents that are relevant to a user's information need.
MAP	Mean Average Precision	A ranking metric that averages precision across relevant results and then across queries.
MRR	Mean Reciprocal Rank	A ranking metric based on the reciprocal rank of the first relevant result for each query.
NDCG	Normalized Discounted Cumulative Gain	A ranking metric that rewards relevant documents appearing higher in the ranked result list.

Continued on next page

Abbreviation	**Meaning**	**Explanation**
OOF	Out-of-fold	Predictions or results produced for samples excluded from the corresponding fitting fold.
RAG	Retrieval-Augmented Generation	An architecture that retrieves external information before or during text generation.
RQ	Research Question	A specific question that the thesis investigates empirically.
SD	Standard Deviation	A statistical measure describing dispersion around the mean.

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Chapter 1

Introduction

1.1 Research Context

Retrieval-Augmented Generation (RAG) uses information retrieved from a document collection to support answer generation by a language model [4]. In legal question answering, the retrieved provisions must be relevant to the question. A fluent answer may still be incorrect if those provisions do not support it. This thesis focuses on retrieval effectiveness. Given a Vietnamese legal question, and a corpus containing legal documents that support the anwser, the task is to rank legal articles or documents so that relevant provisions appear near the top of the results.

Vietnamese legal retrieval presents several challenges. Questions often use everyday language, whereas legal provisions use formal terminology. Articles addressing similar subjects may specify different conditions, obligations, or procedures, so similarity in wording or topic alone does not establish relevance. Corpus completeness also affects retrieval: a ranking model cannot retrieve a provision that is absent from the indexed collection. Incorrect mappings between reference provisions and document identifiers can also affect evaluation results. Retrieval evaluation must therefore consider ranking performance, corpus coverage, and the accuracy of these mappings.

BM25 ranks documents using lexical matches and is relevant to legal retrieval because questions may contain terms, document numbers, or phrases found in the corresponding provisions [5, 6]. This thesis uses BM25's evaluation scores as the baseline. Dense retrieval uses embedding similarity to match questions and documents even when their wording differs. The evaluated embedding models are pretrained BGE-M3 and multilingual-E5-base (E5) [2, 3]. The hybrid configurations, BM25+BGE-M3 and BM25+BGE-M3+E5, combine lexical and dense retrieval scores.

The study evaluates these methods on fixed Vietnamese legal datasets using the available relevance annotations. It compares BM25+BGE-M3 with BM25 and tests whether adding E5 improves BM25+BGE-M3. It also examines whether fine-tuning BGE-M3 improves standalone retrieval and whether any gains remain when the model is integrated into a hybrid. Therefore, this thesis contributes to the understanding of hybrid retrieval strategies in the context of Vietnamese legal information retrieval.

1.2 Research Gap and Approach

Prior research provides a basis for the methods evaluated in this thesis. Rosa et al. report that BM25 is a competitive baseline for legal case retrieval [6]. Chen et al. and Wang et al. present BGE-M3 and multilingual E5 and evaluate their embedding models on multilingual benchmarks [2, 3]. Bruch et al. analyze fusion of lexical and semantic retrieval scores, including the effects of normalization and parameter selection [14]. These findings motivate comparisons between lexical retrieval, dense retrieval, and their combinations.

The scope of that evidence also matters. Legal case retrieval differs from retrieving statutory articles for Vietnamese questions. In heterogeneous retrieval evaluation, BEIR identifies limitations in the generalization of dense retrievers across datasets [17]. Zhan et al. show that training with hard negatives can improve dense retrieval on their evaluated benchmarks [16]; their findings concern dense-retriever training rather than the effect of fine-tuning BGE-M3 within the hybrids evaluated here. Therefore, the literature provides a rationale for testing these methods, while their effectiveness on the selected Vietnamese legal collections remains an empirical question.

The research gap addressed here is the evidence needed for two specific decisions: whether adding E5 improves BM25+BGE-M3, and whether fine-tuning BGE-M3 improves both standalone and hybrid retrieval. The thesis tests these questions under defined corpus conditions and separates model selection from evaluation on a separate test set. This is an empirical contribution rather than a new retrieval algorithm. Chapter 2 reviews the supporting literature, and Chapter 4 specifies the comparisons and evaluation procedures.

The approach has three parts. First, ALQAC and Zalo are evaluated using clean reference text and article text extracted from legal documents collected in raw HTML format. The same questions are used for both text representations, and results are reported separately for each corpus condition. For Zalo, reference text is retained when an extracted article is unavailable.

Second, a Bộ Công an (BCA) dataset is constructed from published public-service questions and legal citations in their answers. The evaluation examines whether the cited provisions are present in the corpus and identifies retrieval failures. The evaluation result is reported separately due to differences in data source from ALQAC and Zalo.

Third, the BGE-M3 embedding model is fine-tuned using legal question article pairs and evaluated in standalone and hybrid retrieval. BM25 and E5 remain unchanged, and fusion weights are held fixed when comparing pretrained and fine-tuned BGE-M3 within each hybrid. The analysis distinguishes development scores from results on test set queries.

1.3 Research Objectives

The main objective is to evaluate hybrid retrieval and fine-tuning of the BGE-M3 embedding model for Vietnamese legal information retrieval. The thesis has four specific objectives:

1. Compare BM25, pretrained BGE-M3, pretrained E5, and BM25+BGE-M3 using the same queries, corpus, and relevance annotations within each evaluation condition. BM25's evaluation scores serve as the baseline.

2. Determine whether adding E5 improves retrieval by comparing BM25+BGE-M3+E5 with BM25+BGE-M3.

3. Evaluate whether fine-tuning BGE-M3 improves standalone retrieval and whether any improvement is retained in BM25+BGE-M3 and BM25+BGE-M3+E5. Model selection uses development data, followed by evaluation on a separate test set. Fusion weights are held fixed when comparing pretrained and fine-tuned BGE-M3 within each hybrid.

4. Construct and evaluate the BCA collection from published public-service questions and legal citations in their answers. Assess the availability of cited provisions in the corpus, the accuracy of mappings between citations and document identifiers, and dependence among questions linked by shared legal references.

Retrieval effectiveness is measured primarily by NDCG@10. Hit@10, MRR@10, and MAP@10 provide additional measures of retrieval performance within the top ten results. Comparisons are reported separately for each dataset and corpus condition to establish the scope of any observed improvement.

1.4 Research Contributions

The thesis contribution is an empirical comparison of BM25, BGE-M3, E5, BM25+BGE-M3, and BM25+BGE-M3+E5 on ALQAC, Zalo, and BCA. The evaluation measures retrieval performance on datasets ALQAC, Zalo, and BCA. The results show that hybrid retrieval improves NDCG@10 over BM25 in most evaluated conditions, while adding E5 provides little to no additional benefit (Tables 5.3 and 5.4).

The second contribution is an evaluation of fine-tuning the BGE-M3 embedding model for standalone and hybrid retrieval. The first experiment selects the fine-tuned model based on hybrid retrieval performance on development data. Development scores improve, but evaluation on a test set does not demonstrate a hybrid improvement. The second experiment selects the model based on standalone dense retrieval performance on development data.

The result shows fine-tuning improves metric NDCG@10 but does not produce a meaningful improvement in either hybrid when fusion weights are the same (Tables 5.11 and 5.12).

The third contribution is the construction and evaluation of the BCA retrieval collection from published public-service questions and legal citations in their answers. The analysis examines the availability of cited provisions in the corpus and dependence among questions linked by shared legal references. The analysis also examines the sortfall failures of the BCA collection (Table 5.5).

1.5 Scope of the Thesis

The thesis evaluates the ranking of legal articles for Vietnamese questions on ALQAC, Zalo, and BCA. It compares BM25, BGE-M3, E5, and their hybrid configurations, and examines the effect of fine-tuning BGE-M3. The evaluation covers retrieval performance, corpus coverage, and the mapping of cited provisions to indexed documents. Answer generation, the quality of legal advice, user satisfaction, the currency of legal documents, and deployment performance are outside its scope.

The evaluation uses Vietnamese questions and legal texts. BGE-M3 and E5 are multilingual models, but the experiments do not assess retrieval in other languages or legal systems. Conclusions are limited to the evaluated datasets, corpus conditions, model configurations, and fusion weights. Applying the findings to other collections or downstream RAG systems requires further evaluation.

For BCA dataset the data is collected from publicly available questions and answers on the Ministry of Public Security's website on the date of collection. Many questions are refrencing private legal documents, which is inaccessible to the public. The evaluation does not attempt to access or retrieve any private legal documents, and the analysis is limited to publicly available information.

1.6 Thesis Organization

Chapter 1: Introduction presents Vietnamese legal information retrieval and the research gap addressed by the thesis. It states the research objectives, contributions, and scope.

Chapter 2: Literature Review reviews prior work on legal retrieval, BM25, multilingual embedding models, hybrid retrieval, and fine-tuning. It also discusses retrieval benchmarks and evaluation methods to establish the basis for the thesis's experimental comparisons.

Chapter 3: Theoretical Background explains lexical and dense retrieval, score normalization and fusion, and contrastive training. It defines relevance, evaluation metrics,

and the principles used to account for dependence among queries when interpreting results.

Chapter 4: Research Methodology describes dataset and corpus construction, including the BCA collection, and the retrieval pipeline. It specifies fusion weight selection, BGE-M3 fine-tuning, and the procedures for model selection and evaluation.

Chapter 5: Experimental Evaluation reports the retrieval comparisons, the effect of adding E5, and the effect of fine-tuning BGE-M3 and hybrid retrieval. It also analyzes BCA corpus coverage, dependence among questions linked by shared legal references, and retrieval failures.

Chapter 6: Conclusion and Future Work summarizes the findings and contributions in relation to the research objectives. It discusses the limitations of the evaluation and outlines directions for future work.

Chapter 2

Literature Review

This chapter reviews prior work on legal retrieval, multilingual embeddings, and hybrid fusion. It identifies the research gap addressed by this thesis and positions its contributions relative to the literature.

Figure 2.1 shows the position of retrieval in a Retrieval-Augmented Generation (RAG) system. The retrieval component searches the document sources and supplies information to the generator to support answer generation.

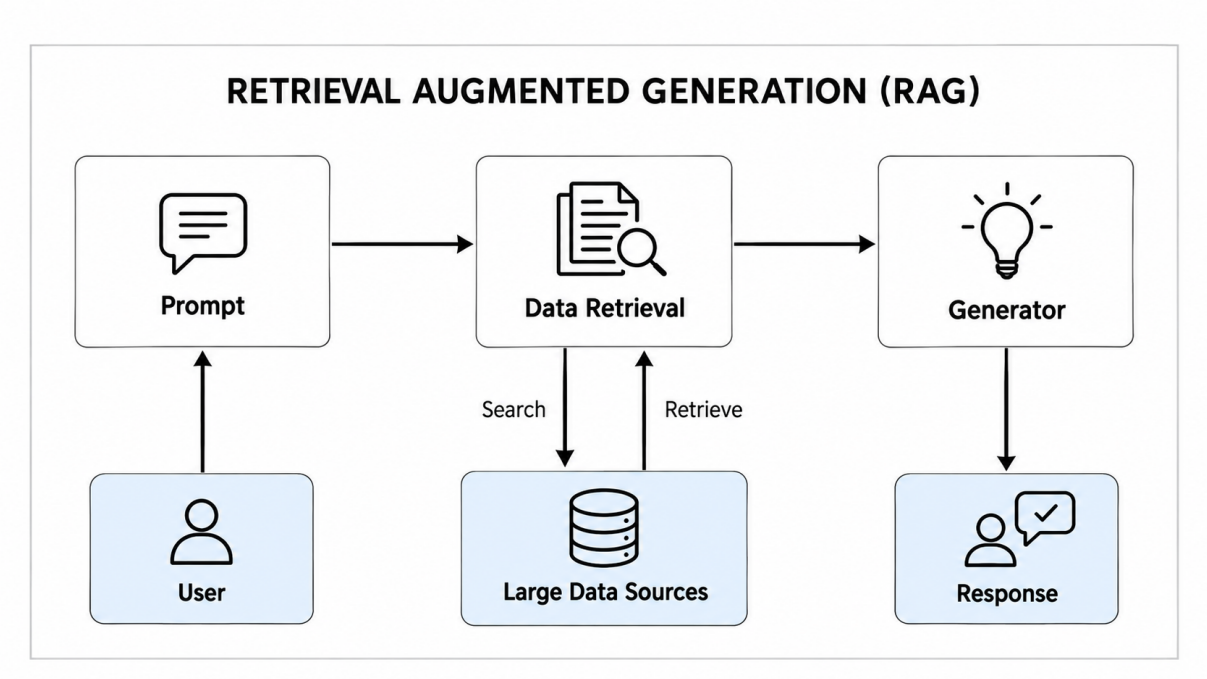


Figure 2.1: Position of retrieval in a Retrieval-Augmented Generation (RAG) system.

2.1 Legal Retrieval and Evidence

Legal information retrieval identifies and ranks legal materials relevant to a query. The retrieval unit may be a judgment, statute, article, or passage. The Competition on Legal Information Extraction and Entailment (COLIEE) separates retrieval tasks from entailment tasks, which assess whether legal texts support a given statement. Its case tasks use Canadian case law, while its statute tasks use Japanese bar-examination questions [7]. Comparisons with Vietnamese article retrieval must therefore account for differences in

the legal system, retrieval unit, relevance annotations, and evaluation metrics.

The Automated Legal Question Answering Competition (ALQAC) 2024 provides manually annotated Vietnamese legal questions and defines separate tasks for legal document retrieval and question answering. In the retrieval task, articles are relevant when they can be used to answer the question [8]. This thesis uses the available relevance annotations to assess how highly the recorded relevant articles are ranked.

Similarity in wording or topic alone does not establish relevance to a legal question. For example, a question about an application deadline may match an article concerning a different stage of the same procedure. In this hypothetical case, the article discusses the same topic but does not provide the required deadline. A relevant provision may also use different terms from those in the question. These possibilities motivate comparing lexical and dense retrieval using the same queries and relevance annotations.

The retrieval unit must also match the relevance annotations. An article may contain rules, exceptions, and references to other provisions, so splitting it into passages changes the context available to the retriever. When the annotations identify articles, retrieved passages must be mapped to their source articles before ranking performance is evaluated. Otherwise, passage retrieval and article retrieval would be assessed using different units.

Text preprocessing is another consideration in Vietnamese retrieval. PhoBERT uses word segmentation before subword tokenization and is evaluated on Vietnamese language-processing tasks [9]. Its preprocessing illustrates a model-specific requirement; its results do not establish that word segmentation improves the retrievers used in this thesis. For lexical retrieval, splitting text at spaces and segmenting it into words produce different vocabularies and term counts. For dense retrieval, preprocessing should follow the embedding model's documented requirements.

In the RAG workflow shown in Figure 2.1, retrieval supplies information to the generator. Lewis et al. combine a language model with retrieved passages for knowledge-intensive tasks [4]. This thesis evaluates the retrieval component by measuring article ranking against the available relevance annotations. Therefore, its results establish retrieval performance within those annotations, while any improvement in the accuracy of generated legal answers requires a separate evaluation.

2.2 Lexical Retrieval and BM25

Lexical retrieval ranks documents by matching query terms with indexed text. BM25 is a probabilistic ranking model that combines term frequency and document frequency with document length normalization [5]. Its scores therefore depend on both the terms in a document and their distribution in the indexed corpus.

A BM25 submission achieved second place in the legal case retrieval task of COLIEE 2021 [6]. This finding provides empirical support for using BM25 as a baseline in legal retrieval. Since that task concerns case law, its results do not establish BM25's effectiveness for Vietnamese legal article retrieval. The latter requires evaluation on the Vietnamese collections studied here.

This thesis compares dense and hybrid retrieval with BM25 using the same queries, corpus, and relevance annotations within each evaluation condition. BM25's evaluation scores serve as the baseline. Section 3.2 explains the scoring formula, and Section 4.5 specifies the preprocessing and retrieval implementation used in the experiments.

2.3 Dense Retrieval and Multilingual Embeddings

Dense retrieval represents queries and documents as embeddings and ranks documents by vector similarity. Sentence-BERT produces sentence embeddings that can be compared using cosine similarity [10]. Dense Passage Retrieval trains separate query and passage encoders for open-domain question answering [11]. Independent encoding allows document embeddings to be computed before a query is submitted and reused during retrieval.

BGE-M3 supports multilingual retrieval through dense, sparse, and multiple-vector representations [2]. Multilingual E5 provides embedding models in small, base, and large variants, trained using multilingual text pairs and labeled datasets [3]. This thesis evaluates BGE-M3's dense representations and multilingual-E5-base (E5). The results therefore apply to these configurations, rather than to every retrieval mode or model variant described in the source papers.

Single-vector retrieval represents each query and document with one vector. ColBERT instead retains contextualized token embeddings and computes relevance through interactions between query and document tokens after encoding [12]. This approach provides an alternative to single-vector retrieval. ColBERT is included here as related work and is not evaluated in the experiments.

Multilingual-E5-base is included to test whether it complements BGE-M3 in semantic retrieval. The models use different training procedures and embedding methods [2, 3]. The hypothesis is that these differences may produce complementary rankings, allowing E5 to improve retrieval for some questions whose relevant articles are ranked poorly by BGE-M3. Whether this improves hybrid retrieval is tested by comparing BM25+BGE-M3+E5 with BM25+BGE-M3.

The experiments also compare pretrained BGE-M3 and E5 with BM25 to assess standalone retrieval on the Vietnamese legal collections, and BM25+BGE-M3 with BM25 to assess hybrid retrieval. These comparisons distinguish the effect of combining lexical and dense retrieval from the effect of adding E5 to an existing hybrid.

2.4 Hybrid Retrieval and Score Fusion

Hybrid retrieval combines the outputs of different retrievers to produce a single ranking. Reciprocal Rank Fusion combines rankings using document positions rather than the original retrieval scores [13]. Weighted score fusion instead combines the component scores according to assigned weights. An analysis of lexical and semantic retrieval examines how score normalization and parameter selection affect fusion performance [14]. These findings motivate specifying both the normalization method and the procedure for selecting fusion weights.

Example of Score Fusion. Consider the legal question:

> *"Is an employee entitled to severance allowance after terminating an employment contract?"*

BM25 and BGE-M3 independently assign retrieval scores to candidate legal articles. Since their scores have different scales, each model's scores are normalized separately over the indexed corpus C. For model m, Min-Max normalization is defined as

$$s'_m(d) = \frac{s_m(d) - \min_{d' \in C} s_m(d')}{\max_{d' \in C} s_m(d') - \min_{d' \in C} s_m(d')},$$

assuming the maximum and minimum scores differ. The normalized scores are then combined using weights of 0.4 for BM25 and 0.6 for BGE-M3:

$$S(d) = 0.4 s'_{\text{BM25}}(d) + 0.6 s'_{\text{BGE-M3}}(d).$$

Table 2.1 shows hypothetical normalized scores for three candidate articles from the larger corpus. The article descriptions and weights are used only to illustrate the calculation.

Table 2.1: Illustrative BM25+BGE-M3 score fusion for three legal articles.

Legal article	**BM25**	**BGE-M3**	**Fused score**	**Final rank**
A: Severance allowance	0.90	0.95	**0.93**	**1**
B: Unemployment benefits	0.85	0.60	0.70	2
C: Contract termination	0.40	0.80	0.64	3

For Article A, the fused score is

$$S(A) = 0.4 \times 0.90 + 0.6 \times 0.95 = 0.93.$$

The articles are ranked by fused score in descending order. Article A ranks first because both models assign it high scores. The example illustrates how fusion combines lexical

matching from BM25 and embedding similarity from BGE-M3 into a single ranking score. It demonstrates the calculation, rather than establishing which provision legally answers the question.

The hypothesis is that lexical and dense retrieval may provide complementary rankings. A relevant article ranked poorly by one retriever may be ranked highly by another. Adding E5 to BM25+BGE-M3 tests whether another dense retriever improves a system that already combines lexical and semantic information. Its standalone performance does not establish that additional benefit.

This thesis uses normalized weighted score fusion and compares BM25+BGE-M3 with BM25+BGE-M3+E5. The weight search includes configurations that assign E5 a weight of zero, so adding E5 expands the available configurations without requiring its use. A higher score on the data used to select weights therefore does not establish an improvement on separate evaluation queries. The selected weights must be evaluated on queries excluded from weight selection.

The evaluation reports the effect of adding E5 separately for each dataset and corpus condition. Section 4.6 specifies score normalization and fusion, and Section 4.8 describes weight selection and evaluation. Reciprocal Rank Fusion is discussed as related work but is not evaluated in the experiments.

2.5 Fine-Tuning for Legal Retrieval and Hard Negatives

The thesis examines two fine-tuning experiments. The first selects the model based on hybrid retrieval performance on development data, followed by evaluation on a separate test set. The second selects the model based on standalone dense retrieval performance and then evaluates it in BM25+BGE-M3 and BM25+BGE-M3+E5. Section 4.7 describes the training procedures, and Section 5.6 reports the results and evaluation limitations.

Contrastive training teaches a retrieval model to give relevant passages higher similarity scores than passages labeled as irrelevant [11]. This thesis uses legal questions and articles to fine-tune BGE-M3 in this way.

Hard negatives are passages that a retriever ranks highly even though they are labeled as irrelevant. They help train the model to distinguish relevant passages from similar but incorrect matches. Experiments show that mixing hard negatives with randomly selected negatives, or selecting new hard negatives during training, can improve dense retrieval [16].

The hybrid comparisons replace pretrained BGE-M3 with fine-tuned BGE-M3 while keeping BM25, E5, and fusion weights unchanged. BGE-M3 is evaluated both on its own and within each hybrid. This tests whether fine-tuning improves standalone retrieval and

whether that improvement is retained when its scores are combined with those of the other retrievers.

2.6 Research Gap and Positioning

BGE-M3 and multilingual E5 provide pretrained embedding models for multilingual retrieval [2, 3]. Research on lexical and semantic score fusion examines the effects of normalization and parameter selection [14], while hard-negative training strategies have improved dense retrieval on benchmark datasets [16]. These studies provide a basis for evaluating multilingual embeddings, score fusion, and embedding model fine-tuning on Vietnamese legal collections.

The research gap addressed by this thesis is the evidence needed to assess these methods under the selected Vietnamese legal datasets and corpus conditions. The evaluation distinguishes improvements over BM25 from improvements due to adding E5 to BM25+BGE-M3. It also tests whether fine-tuning BGE-M3 improves standalone retrieval and whether any improvement is retained in the hybrid systems. Model selection on development data is reported separately from evaluation on a separate test set.

The comparisons use clean reference text and article text extracted from legal documents in raw HTML format.

The contribution is a controlled empirical comparison of established retrieval methods, together with the construction and evaluation of the BCA collection. The BCA analysis examines the availability of cited provisions and dependence among questions linked by shared legal references. Chapter 4 specifies the datasets and evaluation procedures, and Chapter 5 reports the results within those conditions.

2.7 Chapter Summary

This chapter reviewed legal retrieval, BM25, multilingual embeddings, score fusion, and fine-tuning with hard negatives. The reviewed studies motivate evaluating these methods on Vietnamese legal collections, with separate comparisons for improvements over BM25, the contribution of E5, and the effect of BGE-M3 fine-tuning in standalone and hybrid retrieval. Chapter 3 presents the theoretical background and evaluation metrics used in these comparisons.

Chapter 3

Theoretical Background

This chapter presents the theoretical foundations of retrieval models, score fusion, model training, and evaluation metrics used in the thesis. Chapter 4 describes how these methods are implemented and evaluated.

3.1 Queries, Articles, and Relevance

Information retrieval ranks indexed documents according to their relevance to a query [1]. In this thesis, the query is a Vietnamese legal question, and the retrieval unit is a legal article.

The corpus contains laws (Luật), decrees (Nghị định), and circulars (Thông tư). A document may be organized into parts (phần), chapters (chương), sections (mục), subsections (tiểu mục), articles (điều), clauses (khoản), and points (điểm), although not every document uses all these levels [21]. Each article belongs to one legal document and may contain several clauses and points.

Chunking divides document text into units for indexing and retrieval. The choice of unit can affect retrieval performance [22]. This thesis uses articles as the retrieval units. For documents collected in raw HTML format, the extracted text is separated at article boundaries. Each article record contains its text, document identifier, and article identifier. Clauses and points are kept within their source article. Section 4.2 describes corpus construction and the Zalo reference-text fallback when extraction is unavailable.

Chunking can affect recall because it determines the content available to the retriever before encoding. If a chunk incorrectly combines three articles, one vector represents their combined content. This may make a specific provision harder to retrieve and associate text with the wrong article identifier. Improving the encoder does not repair missing text or incorrect identifiers in the corpus. Article boundaries and the match between text and identifiers must therefore be checked when investigating retrieval errors.

Example of an article-level record. The following example contains two article excerpts from the supplied sample for Decree No. 340/2025/NĐ-CP. The title and article texts are shortened for display, with ellipses marking omitted text. The example illustrates the record format; the complete sample retains each article's clauses and points.

```
{
  "law_id": "340/2025/NĐ-CP",
  "title": "NGHỊ ĐỊNH SỐ 340/2025/NĐ-CP ...",
  "url": "https://vbpl.vn/van-ban/chi-tiet/nghi-dinh-so-340-2025-nd-cp-quy-dinh-ve-xu
  ↪  -phat-vi-pham-hanh-chinh-trong-linh-vuc-tien-te-va-ngan-hang--187101",
  "articles": [
    {
      "id": "1",
      "text": "Điều 1. Phạm vi điều chỉnh\n1. Nghị định này quy định các hành vi vi
      ↪  phạm hành chính ... trong lĩnh vực tiền tệ và ngân hàng.\n2. Vi phạm hành
      ↪  chính trong lĩnh vực tiền tệ và ngân hàng bao gồm: ..."
    },
    {
      "id": "2",
      "text": "Điều 2. Đối tượng bị xử phạt vi phạm hành chính\n1. Nghị định này áp
      ↪  dụng đối với tổ chức, cá nhân có hành vi vi phạm hành chính trong lĩnh vực
      ↪  tiền tệ và ngân hàng.\n2. Tổ chức quy định tại khoản 1 Điều này gồm: ..."
    }
  ]
}
```

The fields `law_id`, `title`, and `url` describe the legal document. Each entry in `articles` is a separate chunk with its own `id` and `text`. Article 1 and Article 2 therefore receive separate embeddings. Together, `law_id` and the article `id` identify each retrieval unit.

Relevance targets identify the documents or articles recorded as relevant to each query. ALQAC and Zalo use the available dataset annotations. For BCA, the question is used as the query, and citations in its published answer are extracted to form relevance targets. Document names and article numbers are normalized and matched to corpus identifiers. The answer text is used for target extraction and is not supplied to the retriever. Table 4.2 illustrates this process.

An article-level target specifies a document and an article within it. A law-level target specifies only the document. In the dataset, "law-level" also covers decrees and circulars. Duplicate targets are removed, and each target receives evaluation credit at most once. Retrieving several articles from one document therefore does not multiply credit for a single law-level target. Section 3.6 defines the matching rules and evaluation metrics.

Retrieval returns a list of article identifiers ordered by retrieval score. These identifiers link the results to the article text and source document. A RAG generator can use the retrieved text to produce an answer, while this thesis evaluates the article ranking. Corpus coverage measures target availability, and ranking performance measures target positions.

3.2 Lexical Scoring with BM25

3.2.1 Term Frequency and Inverse Document Frequency

Term frequency and inverse document frequency (TF-IDF) are combined to weight indexed terms. A term receives a higher weight when it occurs frequently in an article but appears in relatively few documents across the corpus [1]. This illustrates how lexical weighting combines term occurrence with term rarity. The experiments use BM25 as the lexical baseline; TF-IDF is included here as background.

3.2.2 BM25

BM25 is a popular lexical scoring function that combines term frequency, inverse document frequency, and document length normalization. For a query q and an article d, a common scoring formula sums the contributions of query terms t [5]:

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t) \frac{f(t,d)(k_1+1)}{f(t,d) + k_1\left(1 - b + b|d|/\operatorname{avgdl}\right)}.$$

The quantities in the formula are:

- $f(t,d)$: the number of occurrences of term t in article d.
- $|d|$: the article length, measured in indexed tokens.
- avgdl: the average article length in the indexed corpus.
- $\mathrm{IDF}(t)$: the inverse document frequency weight of term t.

The parameter k_1 controls term frequency saturation: repeated occurrences of a term contribute progressively less to its score. The parameter b, between zero and one, controls document length normalization. Setting $b = 0$ removes length normalization, while $b = 1$ applies full length normalization [1].

BM25 scores depend on the indexed text. Text extraction can change term counts, article lengths, and corpus statistics, so different corpus representations may produce different scores. A query term absent from an article contributes zero to its BM25 score. Section 4.5 describes the lexical preprocessing used in the experiments.

BM25 relies on matching indexed terms. A relevant article expressed using different terms from the query may therefore receive a low score. The thesis compares BM25 with dense and hybrid retrieval to evaluate whether embedding similarity improves the ranking. BM25 scores determine article order, while evaluation metrics such as NDCG@10 measure ranking quality against the relevance annotations. The baseline is BM25's evaluation performance on the target collections.

3.3 Dense Retrieval and Vector Similarity

An embedding is a numerical vector that represents text. In a bi-encoder architecture, the query and article are encoded separately. Dense retrieval then ranks articles by the similarity between their embeddings and the query embedding. Sentence-BERT produces sentence embeddings for similarity comparison [10]. Dense Passage Retrieval trains query and passage encoders to retrieve relevant passages [11]. Article embeddings can be computed in advance and reused across queries.

Figure 3.1 illustrates separate query and document encoding, similarity scoring, and ranking.

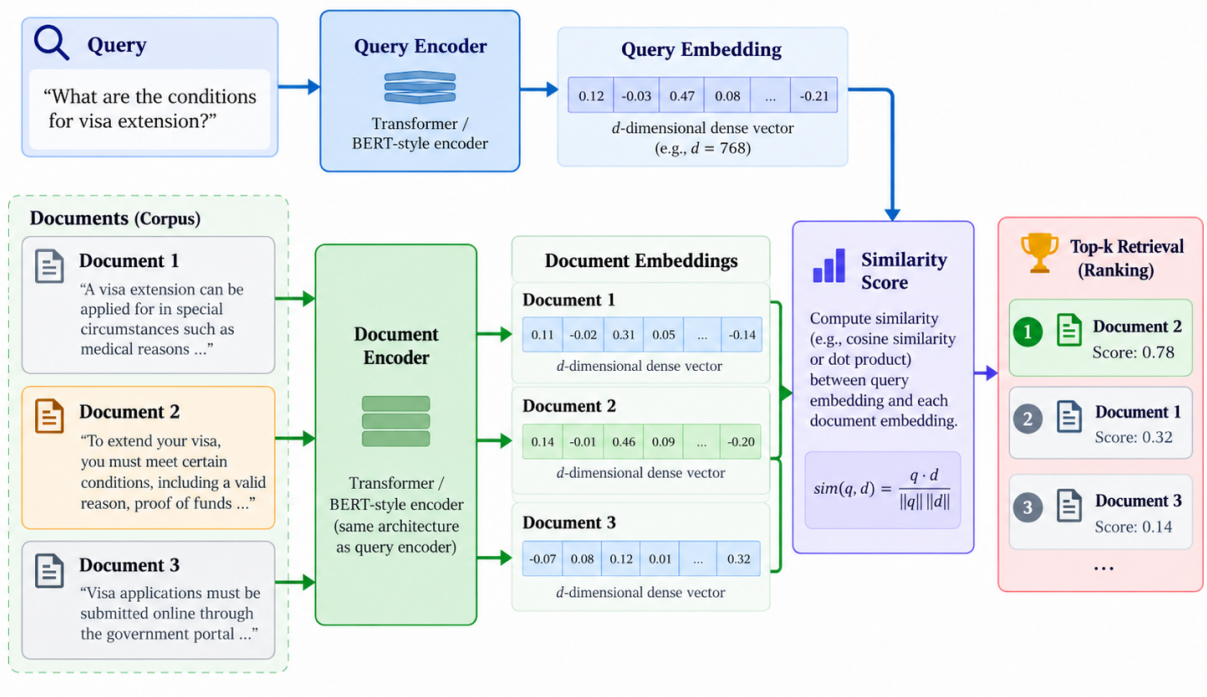


Figure 3.1: Illustrative examples of dense retrieval workflow. The query and document are encoded separately into embeddings, which are then compared using a similarity measure to rank the documents.

For a query q and an article d, u and v are their respective embedding vectors. Two common similarity measures are the dot product and cosine similarity:

$$s_{\text{dot}}(q, d) = u^\top v, \qquad s_{\cos}(q, d) = \frac{u^\top v}{\|u\|_2 \|v\|_2}.$$

The dot product $u^\top v$ sums the products of corresponding vector components. The quantities $\|u\|_2$ and $\|v\|_2$ are the vector lengths. Cosine similarity divides the dot product by these lengths and is defined for nonzero vectors. When both vectors are normalized to length one, the dot product equals cosine similarity.

A pooling operation produces one embedding from the model's token representations.

Input longer than the selected token limit is truncated, so the embedding may omit content that remains in the stored article. ColBERT retains multiple token embeddings for matching [12]. This thesis uses single-vector dense retrieval.

3.3.1 BGE-M3

BGE-M3 has three main features:

1. **Multi-Functionality**

   BGE-M3 supports three retrieval methods [24]:

   - **Dense Retrieval:** Uses dense vector representations to capture semantic meaning.
   - **Sparse Retrieval:** Uses sparse representations to match important keywords and lexical features.
   - **Multi-Vector Retrieval:** Uses multiple vectors with late interaction mechanisms, similar to ColBERT, to capture relevance through comparisons between query and document tokens.

2. **Multi-Linguality**

   - Supports more than 100 languages [2].
   - Enables cross-lingual retrieval, allowing a query to retrieve documents written in a different language [2].

3. **Multi-Granularity**

   - Handles text inputs of varying lengths, from short sentences to long documents [24].
   - Supports input sequences of up to 8,192 tokens for encoding short passages and longer documents [24].

This thesis uses BGE-M3's dense representations for retrieval and fine-tuning. The baseline experiments use a maximum input length of 512 tokens (Table 4.3). The hybrid systems combine BM25 scores with BGE-M3 dense similarity scores. BGE-M3's sparse and multiple-vector modes are outside the evaluated configurations.

3.3.2 Multilingual-E5-base

Multilingual E5 models are pretrained on multilingual text pairs and then fine-tuned on labeled datasets [3]. This thesis uses multilingual-E5-base, released as

`intfloat/multilingual-e5-base`. The model represents each input text as a vector with 768 components [23]. Its weights remain unchanged throughout the experiments.

E5 is included to test whether combining two dense retrievers improves hybrid retrieval. Its contribution is measured by comparing BM25+BGE-M3+E5 with BM25+BGE-M3 using the same queries, corpus, and relevance annotations within each evaluation condition.

3.4 Score Normalization and combination weights

3.4.1 Hybrid Retrieval

Hybrid retrieval combines scores or rankings from different retrievers. The aim is to improve article ranking by combining lexical matches with embedding similarity. The result depends on how the scores are normalized and combined [14]. This thesis evaluates BM25+BGE-M3 and BM25+BGE-M3+E5.

Scores can be combined for all articles in the corpus or only for candidates selected by the retrievers. In the first approach, each retriever scores every indexed article before the scores are combined. In the second, each retriever first selects its top candidates, and fusion ranks only the articles in those lists. An article missing from every candidate list cannot appear in the final results, regardless of the weights used. This thesis uses the first approach: each retriever scores the full corpus before fusion, as described in Section 4.5.

3.4.2 Score Normalization

The retrievers produce raw scores on different scales:

- **BM25:** The score depends on query term matches, term frequency, inverse document frequency, and article length. It has no fixed numerical range and is not normalized to $[0, 1]$. Its values depend on the query, corpus, and inverse document frequency convention [5].
- **BGE-M3:** The pipeline normalizes query and article embeddings to length one and computes their dot product. For nonzero vectors, this equals cosine similarity, whose theoretical range is $[-1, 1]$.
- **Multilingual-E5-base:** The pipeline uses the same normalized-vector dot product, so its theoretical score range is also $[-1, 1]$. Its scores are computed from E5 embeddings rather than BGE-M3 embeddings.

Section 4.5 specifies the score calculation used in the experiments. Although BGE-M3 and E5 share the same theoretical range, their scores can have different distributions across

articles. A raw score from one retriever is therefore not directly interchangeable with a score from another.

Vector normalization and score normalization serve different purposes. Vector normalization gives each embedding a length of one. Min-Max score normalization rescales each retriever's scores across the corpus for a particular query before weighted fusion [14]. For raw score $s_j(q, d)$ from retriever j, the form used in this thesis is

$$\widetilde{s}_j(q, d) = \frac{s_j(q, d) - \min_{x \in C} s_j(q, x)}{\max\left(\max_{x \in C} s_j(q, x) - \min_{x \in C} s_j(q, x), \epsilon\right)},$$

where C is the indexed corpus and $\epsilon > 0$ prevents division by zero. The minimum and maximum are computed separately for each query and retriever. The resulting scores lie in $[0, 1]$. If all articles receive the same raw score, their normalized scores are zero. Normalization preserves each retriever's score ordering; it changes the scale used for fusion. These scores are used to rank articles and are distinct from evaluation metrics such as normalized discounted cumulative gain. Section 4.6 specifies the implemented normalization procedure.

An alternative is z-score normalization:

$$Z_j(q, d) = \frac{s_j(q, d) - \mu_j(q)}{\sigma_j(q)},$$

where $\mu_j(q)$ and $\sigma_j(q) > 0$ are the mean and standard deviation of the retriever's scores over the corpus. Z-scores can be negative and are not restricted to the range $[0, 1]$. This method is included as background; the reported experiments use Min-Max normalization. Both methods depend on the articles included in normalization, so the normalization set must be stated.

3.4.3 Weighted Score Fusion

Weighted score fusion adds the normalized retrieval scores after multiplying each by a weight [14]. The weights control how much each retriever contributes to the combined score. For a query q and an article d, $\widetilde{s}_b$, $\widetilde{s}_g$, and $\widetilde{s}_e$ denote the normalized scores from BM25, BGE-M3, and E5, respectively.

For BM25+BGE-M3, the combined score is

$$S_{BG}(q, d) = w_b \widetilde{s}_b(q, d) + (1 - w_b)\widetilde{s}_g(q, d), \qquad 0 \leq w_b \leq 1,$$

Here, w_b is the BM25 weight, and $1 - w_b$ is the BGE-M3 weight.

For BM25+BGE-M3+E5, the score becomes

$$S_{BGE}(q,d) = w_b\widetilde{s}_b(q,d) + w_g\widetilde{s}_g(q,d) + w_e\widetilde{s}_e(q,d),$$

The weights w_b, w_g, and w_e correspond to BM25, BGE-M3, and E5. They are nonnegative and satisfy $w_b + w_g + w_e = 1$. Articles are ranked by the combined score in descending order. Section 4.6 specifies the fusion procedure used in the experiments.

When $w_e = 0$, E5 does not contribute to the combined score. Setting $w_g = 1 - w_b$ then reproduces BM25+BGE-M3, provided the corpus, retrieval scores, and normalization are unchanged. The weight search can therefore select a configuration that omits E5. A higher score on the data used to select weights does not establish an improvement on a separate test set. Therefore, E5's contribution is assessed by comparing the selected BM25+BGE-M3+E5 and BM25+BGE-M3 configurations on a separate test set.

3.5 Fine-Tuning and Negative Sampling

Pretraining on legal corpora develops language representations, as investigated in LEGAL-BERT [15]. Retrieval fine-tuning instead uses labeled questions and relevant passages to train their similarity scores [11]. This section explains the training objectives and negative examples used to fine-tune the BGE-M3 embedding model.

3.5.1 Contrastive Learning

BGE-M3 is fine-tuned to give relevant articles higher similarity scores than articles used as negative training examples. For a question q, d^+ is one article labeled as relevant, and $N(q)$ is the set of negative training examples. The similarity between a question and an article is denoted by $s(q,d)$. A single-positive contrastive loss is [11]

$$\mathcal{L}(q) = -\log \frac{\exp(s(q,d^+)/\tau)}{\exp(s(q,d^+)/\tau) + \sum_{d^-\in N(q)} \exp(s(q,d^-)/\tau)}.$$

The loss decreases when the relevant article receives a higher score relative to the negative examples. The temperature $\tau > 0$ scales the similarity scores within the loss. Smaller values increase the effect of score differences.

A question may have several relevant articles among its training candidates. The multi-positive objective used in this thesis sums their exponentiated scores in the numerator and sums the exponentiated scores of all candidates in the denominator. This allows several relevant articles to contribute to the objective for one question.

The experiment that selects the model based on hybrid retrieval performance uses a

single-positive loss. The experiment that selects the model based on standalone dense retrieval performance uses a multi-positive loss. Section 4.7 specifies the candidates and loss used in each experiment. A lower training loss must be followed by evaluation on a separate test set to assess retrieval performance.

3.5.2 Negative Sampling

Negative sampling selects articles that the model is trained to rank below the known relevant articles. Three common methods differ in how the examples are selected [11, 16]:

- **Random negatives:** Articles selected at random from the corpus.
- **In-batch negatives:** Articles from other training examples processed in the same batch. An article relevant to another question may be treated as a negative for the current question unless it is also identified as relevant to that question.
- **Hard negatives:** Articles ranked highly by a retriever but selected as negatives because they have no recorded positive label for the question.

An article without a positive label may still be relevant. Excluding known relevant articles from negative candidates reduces contradictory training labels but does not resolve missing annotations. The training procedure must also state how it treats other known relevant articles within the same batch. Section 4.7 describes the negative-selection rules and treatment of positive examples in the fine-tuning experiments.

3.5.3 Hard-Negative Mining

Hard-negative mining selects negative training examples from a retriever's rankings [16]. The procedure must specify the retriever, corpus, candidate ranks, and filtering rules so that the training examples can be reproduced.

In the experiment that selects BGE-M3 based on hybrid retrieval performance, negatives are selected from lexical and dense rankings. Known relevant articles, duplicate candidates, unavailable articles, and articles from legal documents reserved for evaluation are excluded. Section 4.7 describes the construction of these training examples.

3.6 Target Matching and Evaluation Metrics

Evaluation compares ranked articles with the relevance targets recorded for each query. An article-level target requires both the legal document identifier and the article identifier to match. A target with an empty article identifier or `toanvan` refers to the whole legal

document, so any article from that document can match it. Each target is counted at most once. Retrieving several articles from the same document therefore does not increase the score for a single whole-document target. When a query has both types of target, the evaluator's target-checking order determines which unmatched target receives credit.

In the following formulas, Q is the set of evaluation queries, and R_q is the set of distinct recorded relevance targets for query q. The value $y_{q,r}$ is one if the article at rank r matches a previously unmatched target, and zero otherwise. The cutoff k specifies how many articles from the top of the ranking are evaluated.

Normalized discounted cumulative gain. Normalized discounted cumulative gain (NDCG) assigns more credit to relevant results at higher ranks and compares the observed ranking with an ideal ranking [18]. The thesis uses NDCG@10 as the primary metric, with binary credit defined by the target-matching rules above:

$$\mathrm{NDCG@10}(q) = \frac{\sum_{r=1}^{10} y_{q,r} / \log_2(r+1)}{\sum_{r=1}^{\min(|R_q|,10)} 1/\log_2(r+1)}.$$

The denominator represents an ideal ranking in which up to ten targets are matched at the highest ranks. Queries without targets receive zero. Targets that cannot be matched to indexed articles remain in the denominator. The score therefore reflects both ranking quality and limitations in corpus coverage or identifier mapping.

Hit and recall. Hit@k is one if at least one target is matched within the first k results, and zero otherwise:

$$\mathrm{Hit@}k(q) = \mathbb{1}\left[\sum_{r=1}^{k} y_{q,r} > 0\right],$$

Here, $\mathbb{1}[\cdot]$ equals one when its condition is satisfied and zero otherwise. Hit is reported at $k = 1, 5, 10$. Recall measures the proportion of recorded targets that are retrieved [1]. Under the matching rules used here, it is calculated for queries with at least one target as

$$\mathrm{Recall@}k(q) = \frac{\sum_{r=1}^{k} y_{q,r}}{|R_q|}.$$

For example, matching one of two targets within the first k results gives Hit@$k = 1$ and Recall@$k = 0.5$. The metrics coincide for queries with one target. The baseline results report Hit, while the fine-tuning comparisons also report Recall@100 (Tables 5.11 and 5.12).

Mean reciprocal rank. Mean reciprocal rank (MRR) measures how early the first target is matched. For a query whose first matching article occurs at rank r_q^*, the reciprocal

rank (RR) at cutoff ten is

$$\mathrm{RR@10}(q) = \begin{cases} 1/r_q^*, & \text{if a target is matched within the first ten results,} \\ 0, & \text{otherwise.} \end{cases}$$

MRR@10 is the mean of these values across queries:

$$\mathrm{MRR@10} = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{RR@10}(q).$$

It considers only the first match, so additional matches do not change the score.

Mean average precision. Average precision (AP) combines precision values at ranks where relevant results are retrieved. Mean average precision (MAP) averages AP across queries [1]. In this thesis, $P_q(r)$ is the proportion of the first r results that match distinct targets. The evaluator computes the metrics at cutoff ten as

$$P_q(r) = \frac{1}{r} \sum_{j=1}^{r} y_{q,j},$$
$$\mathrm{AP@10}(q) = \frac{\sum_{r=1}^{10} y_{q,r} P_q(r)}{\max(1, \min(|R_q|, 10))},$$
$$\mathrm{MAP@10} = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{AP@10}(q).$$

The AP@10 denominator is the number of recorded targets, capped at ten. The lower bound of one avoids division by zero and gives queries without targets a score of zero. This implementation differs from an AP definition that divides by the total number of targets without a cap.

Each aggregate metric gives equal weight to each query and includes queries with zero scores, unless the results explicitly restrict evaluation to queries with available targets. These metrics measure retrieval against the recorded annotations. They do not establish that every legally relevant article has been retrieved or that a generated legal answer is correct.

3.7 Chapter Summary

This chapter explains how BM25 and dense retrieval models score legal articles, how score fusion combines their rankings, and how fine-tuning updates the BGE-M3 embedding model. It also defines the target-matching rules and evaluation metrics used to compare retrieval performance. Chapter 4 describes the experimental setup and evaluation procedures.

Chapter 4

Research Methodology

This chapter describes the research design, datasets, and implementation used to evaluate Vietnamese legal article retrieval. It presents corpus preparation and BCA dataset construction before the retrieval pipeline, score fusion, and BGE-M3 fine-tuning procedures. It then specifies the evaluation design, statistical analysis, experimental environment, and data-use limitations.

4.1 Research Design and Retrieval Task

This thesis uses controlled retrieval experiments to compare lexical retrieval, dense retrieval, and their combination for Vietnamese legal questions. The system receives a question, scores the indexed legal articles, and returns a ranked list. Each article has a legal document identifier and an article identifier. The primary outcome is normalized discounted cumulative gain at rank ten (NDCG@10); the matching rules and evaluation metrics are defined in Section 3.6.

BM25 provides the baseline for retrieval effectiveness. Comparing BM25+BGE-M3 with BM25 assesses the benefit of combining lexical and dense retrieval. Comparing BM25+BGE-M3+E5 with BM25+BGE-M3 assesses the additional benefit of multilingual-E5-base. Dataset and corpus conditions provide separate settings for these comparisons, with the same query population retained within each comparison of corpus versions. The evaluation also distinguishes articles absent from the corpus from errors in mapping target identifiers, since both can affect measured performance.

The fine-tuning experiments compare pretrained and fine-tuned BGE-M3 in standalone dense retrieval and in the hybrid configurations. One experiment selects the model using hybrid retrieval performance on development data; the other uses standalone dense retrieval performance. The hybrid comparisons keep BM25, E5, and the combination weights fixed while replacing BGE-M3 with its fine-tuned version. This design tests whether improvements in standalone retrieval are retained after score fusion. Section 4.7 describes the training procedures, and Section 4.8 specifies the evaluation conditions and their limitations.

Relevance labels support training, configuration selection, and evaluation. At retrieval time, the system uses only the question text and the indexed corpus. Published answers

and recorded relevance targets are not retrieval inputs. The experiments therefore measure recovery of annotated legal articles rather than the correctness of a generated legal answer.

4.2 Datasets and Corpus Preparation

The evaluation uses ALQAC, Zalo, and the BCA collection. For ALQAC and Zalo, it compares a corpus containing the supplied reference article text with a corpus prepared using article text extracted from legal documents collected in raw HTML format. For Zalo, reference text is retained when a corresponding extracted article is unavailable or empty, as explained below. Each comparison uses the same ordered queries and relevance targets, so the corpus versions are evaluated as paired conditions rather than independent datasets.

Table 4.1 summarizes the five baseline evaluation conditions. Queries that share legal references are grouped for selection and evaluation, as explained in Section 4.8.

Table 4.1: Baseline datasets and corpus versions.

Dataset	**Queries**	**Articles**	**Evaluation design**
ALQAC, reference text	811	3,363	Grouped cross-validation using reference article text
ALQAC, extracted text	811	3,381	Same ordered queries, with a separately extracted article corpus
Zalo, reference text	3,196	61,425	Grouped cross-validation using reference article text
Zalo, extracted text with reference fallback	3,196	61,425	Same ordered queries, with extracted text where available and reference text otherwise
BCA	546	14,789	Separate evaluation of the largest linked group and grouped cross-validation over smaller groups

4.2.1 ALQAC and Zalo Datasets

ALQAC and Zalo provide Vietnamese legal questions with annotations identifying the relevant legal articles. The questions serve as retrieval queries, and the annotated references provide the targets used to evaluate the ranked results.

For ALQAC, the baseline evaluation combines the supplied training and private-test questions, giving 811 queries. Its reference corpus contains 3,363 articles.

For Zalo, the baseline evaluation uses 3,196 queries and a reference corpus of 61,425 articles. Each query retains its recorded relevance targets when evaluated against either the reference-text corpus or the extracted-text corpus with reference fallback.

For both datasets, questions linked by shared legal references are kept in the same group. Fusion weights are selected using other groups before evaluation on the excluded group, as described in Section 4.8. Within each dataset, the same queries and relevance targets are used for both corpus versions so that their retrieval results can be compared directly. The corpus preparation procedures are described in the following subsection.

4.2.2 Article Extraction and Identifier Alignment

The reference corpus are organized into individual article records, each containing a legal document identifier, an article identifier, and the article text. For the extracted-text versions, legal documents collected in raw HTML format are converted to text and separated into articles. The retrievers use the resulting article text rather than raw HTML. Document and article identifiers link each retrieved result to its recorded relevance target.

For ALQAC, a document mapping links the reference documents to their HTML sources. Articles are extracted from those sources, with a leading article heading removed where present. The resulting corpus contains 3,381 articles, compared with 3,363 reference articles. Extraction can change article boundaries and identifiers, so a larger corpus does not necessarily contain more of the annotated targets.

For Zalo, preparation follows the reference article order. An extracted article replaces the reference text only if both identifiers match and its cleaned text is nonempty. If no matching extracted article is available or its cleaned text is empty, the reference text is retained. This procedure is called *reference fallback*.

Fallback keeps all 61,425 articles searchable with their original identifiers and order. Removing articles without usable extracted text would also remove retrieval candidates, making it harder to distinguish changes caused by article availability from changes caused by text extraction. The resulting Zalo corpus therefore combines extracted text with retained reference text. Its comparison with the reference corpus evaluates this replacement procedure, not a complete replacement of all reference text.

Identifier normalization is separate from text preprocessing. Document and article identifiers are normalized before matching retrieved articles to relevance targets. Within each corpus version, all retrievers use the same article order so that scores at the same position refer to the same article. Incorrect identifier alignment can prevent a relevant retrieved article from receiving evaluation credit even when its text is present.

4.3 BCA Dataset Construction

The BCA dataset is constructed from public questions and published answers on the Bộ Công an portal. Each question becomes a retrieval query, and legal references extracted from its published answer become the recorded relevance targets. Construction consists of collecting the source records, cleaning their text, extracting legal references, and mapping those references to corpus identifiers. The answer text is used to construct targets and is not supplied to the retriever.

4.3.1 Motivation and Research Value

The first purpose is to extend evaluation to questions submitted through a public-service channel. Competition benchmarks provide defined tasks and evaluation conditions; for example, ALQAC evaluates legal retrieval and question answering using a manually annotated dataset [8]. BCA adds questions concerning residence, identity documents, fire prevention, traffic penalties, and immigration. Citizens may describe their circumstances in everyday language rather than use the terms found in legislation. This motivates examining retrieval on public-service questions alongside the ALQAC and Zalo benchmark comparisons.

The second purpose is to construct a research resource beyond the supplied benchmarks. The collection and reference-extraction pipeline converts published question-answer records into queries and traceable relevance targets. It supports evaluation of BM25, BGE-M3, E5, and their combinations on a separately collected set of questions. The resulting analysis examines performance across collections, including differences in target availability and shared legal references. The resource contribution is the constructed collection and its documented processing procedure.

4.3.2 Data Sources and Record Selection

The collection script requests successive pages of question listings through the portal's application programming interface, then retrieves each question and its published answer. Each record is assigned an identifier derived from the source question identifier. During a repeated collection, records with existing identifiers are skipped to prevent duplication.

Acquisition requires separate listing and detail requests: the listing identifies available questions, while each detail response supplies the question and answer fields. Failed listing requests can interrupt discovery, and failed detail requests can leave individual records uncollected.

The saved collection contains 652 records with distinct question identifiers. Of these, 546 are retained for retrieval evaluation because they contain at least one extracted legal

document reference.

4.3.3 Text Cleaning and Reference Extraction

Preprocessing covers the collected question-answer records, extracted legal articles, and inputs to each retriever. Legal references are then derived from the published answers to construct evaluation targets. The question text remains the retrieval query throughout this process.

Text cleaning and retriever inputs. The preprocessing steps are organized by their role:

1. **Cleaning question-answer records.** The response fields `content` and `answer` contain the question and published answer. Regular expressions replace HTML tags with spaces, remove leading and trailing whitespace, and reduce repeated spaces, tabs, and line breaks to a single space. Unicode Normalization Form C (NFC) standardizes equivalent character sequences in Vietnamese text. The cleaned question is used as the retrieval query, while the cleaned answer is used for reference extraction.

2. **Extracting legal articles.** Legal HTML is converted to text lines using BeautifulSoup. Article boundaries are identified from lines beginning with "Điều" and its alternative spelling, followed by an article number and an optional letter suffix. The leading article marker and number are removed where present.

3. **Preparing BM25 tokens.** Question and article text is converted to lowercase. Punctuation and other characters outside the word-character and whitespace classes are replaced with spaces, and the text is split on whitespace. Vietnamese letters with diacritics are retained. This procedure uses no external Vietnamese word-segmentation library.

4. **Formatting dense inputs.** BGE-M3 receives question and article text without additional prefixes. Multilingual-E5-base receives the prefix `query:` for questions and `passage:` for articles. Both models use a maximum input length of 512 tokens in the baseline experiments. Section 4.5 describes the retrieval models, and Section 4.7 specifies the separate fine-tuning configurations.

Legal reference extraction. The thesis processes the published answer using the following steps:

1. **Identifying articles and documents.** The answer is divided into spans using punctuation and line boundaries. Within each span, the extractor identifies article

numbers, including forms such as “Điều 5”, “Điều 12a”, and “Điều 32”. It also detects document terms such as “Luật”, “Bộ luật”, “Nghị định”, and “Thông tư”.

2. **Filtering ambiguous phrases.** Keywords such as “quy định”, “quyết định”, and “nghị quyết” require a following number to be treated as document references. Ordinary expressions such as “luật sư” and “luật chơi” are excluded. The rules also filter references marked as drafts and remove trailing phrases, incomplete parentheses, and other text outside the detected document name. Document-specific exceptions handle cases not covered by the general rules.

3. **Linking articles to documents.** The extractor first attempts to link an article mention to the following document mention when the intervening words are permitted connectors. If that association fails, it checks the preceding document using a separate connector-word set. This handles both “Điều 9 Nghị định 79/2014/NĐ-CP” and “Nghị định số 79/2014/NĐ-CP tại Điều 9, Điều 15”. A document cited without a specific article is recorded with an empty article identifier, representing a whole-document target.

4. **Removing duplicate references.** Repeated document and article pairs are removed. When two extracted names refer to the same article and one name is contained in a longer version, the more complete name is retained. For example, “Luật Giao thông đường bộ năm 2008” is retained over the shorter “Luật Giao thông đường bộ” when both have been extracted for the same article.

The extracted references are saved with the cleaned question-answer record. Section 4.3.4 describes their mapping to corpus identifiers and the checks for target availability. Evaluation uses the matching rules in Section 3.6: a whole-document target can match any indexed article from that document, and each recorded target receives credit at most once. Figure 4.1 summarizes the construction process and its connection to corpus coverage checks.

4.3.4 Target Mapping and Corpus Coverage

An extracted document name may differ from the identifier used in the corpus. Answers can refer to a document by its common title or an abbreviated number, such as “Nghị định 100”, while the corpus uses a full identifier. Such a mention cannot be resolved reliably from its abbreviated number alone. A stored correction table, `corrected_mappings.json`, records the mapping decisions that link extracted names to document identifiers and, where available, extracted article files. Article identifiers are normalized separately. The resulting document and article identifiers are used to match targets with indexed articles.

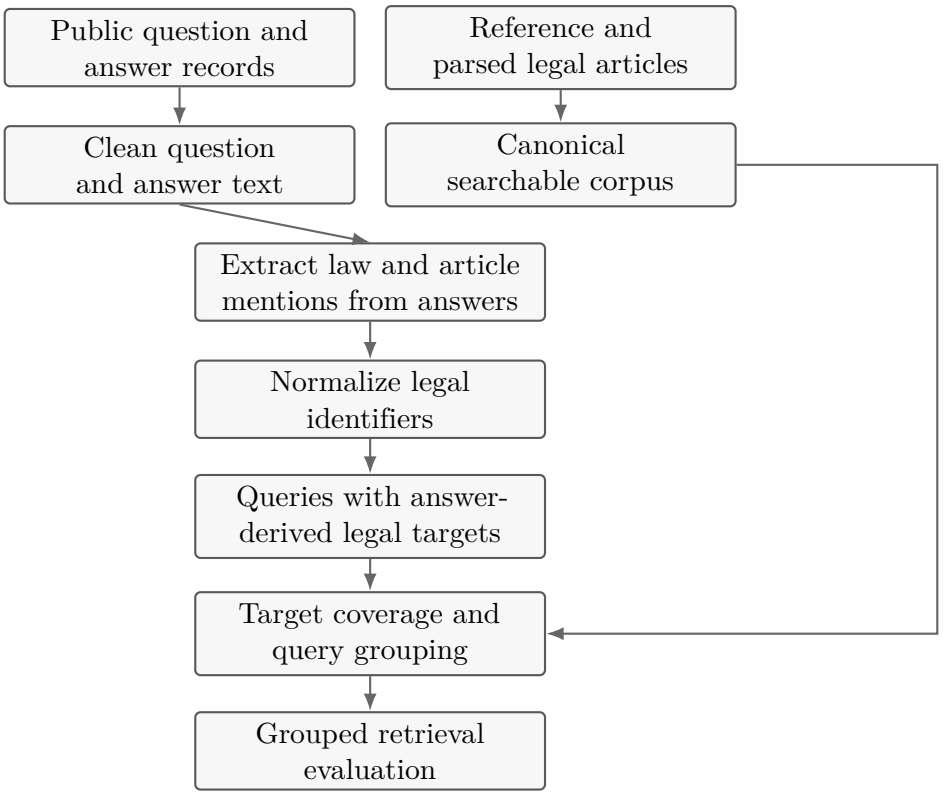


Figure 4.1: Construction of the BCA retrieval collection. Published answers supply reference evidence; only question text is used as the retrieval query.

A supplementary collection script, `crawl_batch_laws.py`, searches for additional legal documents and saves raw HTML, extracted articles, and source mappings when a suitable public source is found. This step expands the searchable corpus, but unsuccessful searches and ambiguous document names can leave references unresolved. Correcting an identifier and obtaining the referenced text are separate requirements.

The searchable corpus starts from reference articles and is extended with article text extracted from legal documents collected in raw HTML format. Duplicate article identifiers within each document and articles with empty text are excluded. The resulting corpus contains 14,789 articles.

Some references to Ministry of Public Security documents remain unresolved after public-source searches. The mapping procedure can assign the placeholder `INTERNAL_DOCUMENT` following an unsuccessful search for a reference containing "BCA". Entries with this marker are excluded from added searchable documents, while their query targets can remain unavailable. The marker records a processing outcome; an unsuccessful search alone does not establish that a document is internal, confidential, or absent from every public legal database.

For each retained question, coverage checks determine whether its targets match indexed articles under the rules in Section 3.6. Questions may have all, some, or none of their targets available. An article absent from the corpus cannot be retrieved. An incorrect

identifier mapping can instead prevent a retrieved article from receiving credit even when the relevant text is present. These cases must be distinguished from a ranking error, where an available target is ranked too low.

The retained questions are grouped by shared legal references for evaluation, as described in Section 4.8. Questions with partially available or unavailable targets remain in the full-collection evaluation. Chapter 5 reports their coverage and retrieval results separately where needed.

4.3.5 Example of Query and Target Construction

Table 4.2 illustrates the conversion of a saved question into a query and relevance targets. Its published answer cites Nghị định số 79/2014/NĐ-CP. The saved extraction contains a whole-document reference and references to Articles 9 and 15. These entries illustrate the stored dataset and do not assert the decree's current applicability.

Table 4.2: Worked BCA question-to-target transformation.

Stage	**Representation**	**Interpretation**
Stored query	"Bộ Công an cho tôi hỏi có quy định bắt buộc nào về các trang thiết bị, dụng cụ phòng cháy chữa cháy đối với nhà chung cư không?"	Question text supplies the retrieval input
Extracted answer references	Nghị định số 79/2014/NĐ-CP, with article identifiers empty, 9, and 15	The saved extraction contains one whole-document reference and two article references
Mapped document identifier	79/2014/NĐ-CP	The extracted document name is mapped to a corpus identifier
Evaluation targets	The normalized document identifier paired with each recorded article identifier	An empty article identifier permits whole-document matching; 9 and 15 require the corresponding articles
Retriever input	Question text and indexed article text	The published answer and recorded targets are not supplied at retrieval time

Whole-document and article-level targets can coexist in one record. Each target receives credit at most once under the matching rules in Section 3.6. The three extracted references in this example are recorded targets, not three independent expert judgments of legal relevance.

4.4 System Overview

The system takes a Vietnamese question and an indexed corpus of legal articles as inputs. BM25 assigns lexical scores, while BGE-M3 and multilingual-E5-base assign dense similarity scores. Each retriever scores every article in the same corpus order. The experiments compare individual retrievers with BM25+BGE-M3 and BM25+BGE-M3+E5. The evaluated pipeline ends at article ranking and does not include reranking or answer generation.

For hybrid retrieval, scores are normalized separately for each query and retriever, then combined by weighted addition. The combination weights are selected on designated queries and fixed before ranking the evaluation queries. The system returns the ten highest-ranked articles with their legal document and article identifiers. Section 4.6 specifies the normalization, combination weights, and ranking procedure.

Evaluation is performed after retrieval by comparing the returned identifiers with the recorded relevance targets. Published answers and relevance targets are not retrieval inputs. Questions linked by shared legal references are assigned to the same evaluation group, as described in Section 4.8. Figure 4.2 shows the baseline scoring and evaluation pipeline. Fine-tuning replaces the BGE-M3 component within this pipeline; Section 4.7 describes the training and model selection procedures.

4.5 Retrieval Models and Implementation

The experiments use BM25 for lexical retrieval and the dense representations of BGE-M3 and multilingual-E5-base for semantic retrieval. All models score the same ordered articles within each corpus version. Preserving article order allows their scores to be combined without changing the associated legal document and article identifiers.

4.5.1 BM25

BM25 scores tokenized question and article text. Preprocessing converts text to lowercase, replaces non-word punctuation with token separators, and splits the resulting text on whitespace. This preprocessing remains fixed in the reported comparisons. The experiments therefore evaluate this BM25 configuration rather than alternative Vietnamese tokenization methods.

4.5.2 BGE-M3 and Multilingual-E5-base

BGE-M3 is the primary dense retriever. Multilingual-E5-base, referred to as E5, provides an additional dense score for the BM25+BGE-M3+E5 configuration. The experiments

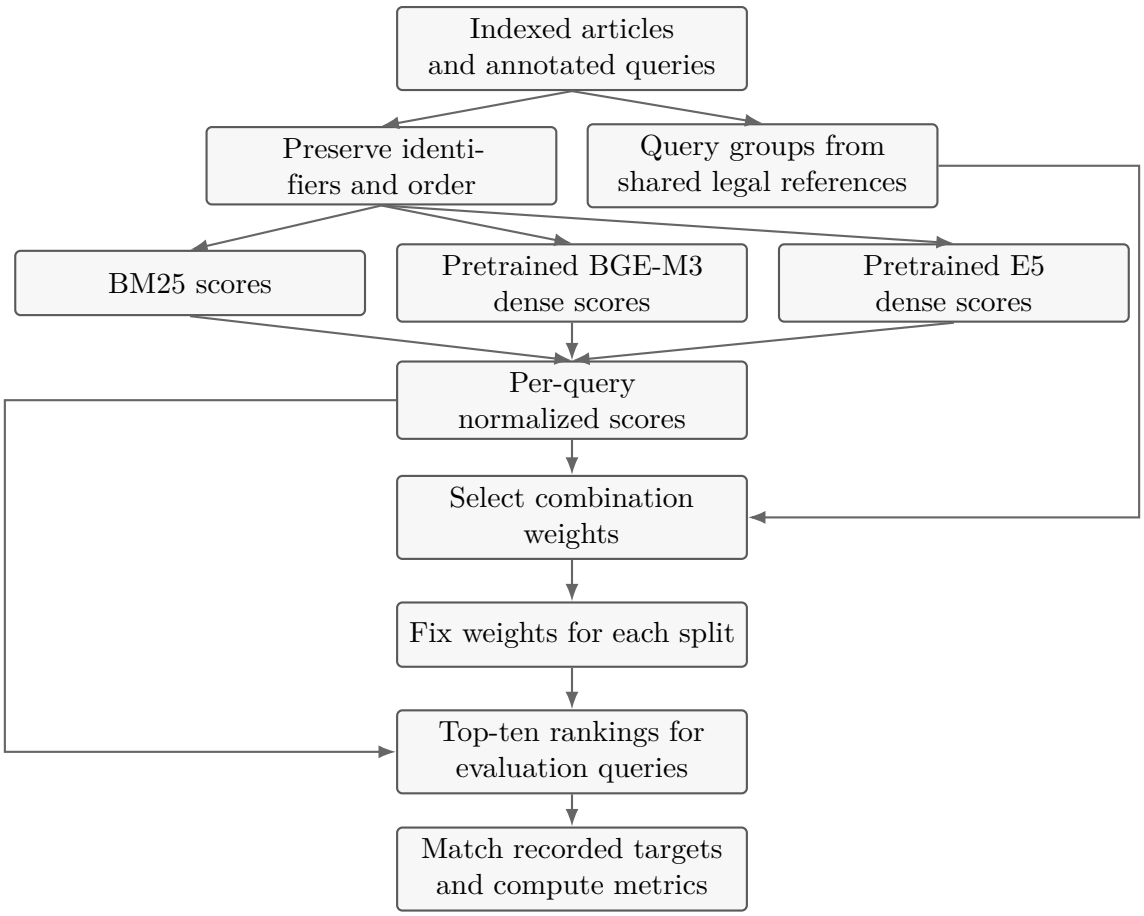


Figure 4.2: Baseline scoring and evaluation pipeline. Scores are computed over the full corpus. Relevance targets are used for grouping, weight selection, and evaluation, but are not supplied to the retrievers.

use BGE-M3's dense representation; its sparse and multi-vector retrieval methods are not evaluated.

Each encoder maps the question and each article to vectors. The implementation divides each vector by its Euclidean norm, with a minimum denominator of 10^{-10}, and computes the dot product of the normalized vectors. For the query vector $\mathbf{u}_q$ and article vector $\mathbf{v}_d$ before normalization, the resulting score is

$$s_{\text{dense}}(q, d) = \frac{\mathbf{u}_q^\top \mathbf{v}_d}{\max(\|\mathbf{u}_q\|_2, 10^{-10}) \max(\|\mathbf{v}_d\|_2, 10^{-10})}.$$

Here, q is the question, d is an indexed article, and $\|\cdot\|_2$ denotes the Euclidean norm. For vectors whose norms exceed the numerical threshold, this calculation is cosine similarity. The threshold prevents division by zero.

Table 4.3 summarizes the pretrained dense models and the baseline input settings. Both encoders use a maximum input length of 512 tokens in the baseline experiments. This experimental limit is distinct from BGE-M3's supported maximum of 8,192 tokens described in Chapter 3. The fine-tuning experiment selected by hybrid retrieval performance uses maximum lengths of 256 tokens for queries and 512 tokens for passages, as specified in Section 4.7.

Table 4.3: Baseline dense-model settings.

Property	**BGE-M3**	**Multilingual-E5-base**
Model family	BGE-M3 multilingual embedding model	E5 multilingual text-embedding model
Embedding dimension	1,024	768
Experimental role	Primary pretrained dense component	Additional pretrained dense component
Baseline maximum input length	512 tokens	512 tokens
Fine-tuning status	Fine-tuned in the experiments in Section 4.7	Pretrained parameters remain fixed

The evaluation scores every indexed article rather than restricting retrieval to candidates from an approximate nearest-neighbor index. Thus, an available target that falls outside the first ten results is excluded by the resulting ranking rather than by a separate candidate filter. This exhaustive scoring design supports comparisons of ranking performance; it does not establish the system's latency in deployment.

4.6 Score Normalization and Fusion

BM25 and dense retrieval produce scores on different scales. Before combining them, the baseline applies Min-Max normalization separately to each retriever's scores for each query. For retriever r, raw score $s_r(q, d)$, and full indexed corpus C, the normalized score is

$$\widetilde{s}_r(q, d) = \frac{s_r(q, d) - \min_{d' \in C} s_r(q, d')}{\max\left(\max_{d' \in C} s_r(q, d') - \min_{d' \in C} s_r(q, d'), \epsilon\right)}.$$

Here, ϵ is a small positive constant that prevents division by zero. If a retriever assigns the same score to every article for a query, all its normalized scores are zero. Normalization uses the full corpus rather than a shortlist of retrieved candidates, so every indexed article can enter the final ranking.

The BM25+BGE-M3 score is

$$S_{bg}(q, d) = w_b \widetilde{s}_b(q, d) + (1 - w_b)\widetilde{s}_g(q, d),$$

where b denotes BM25, g denotes BGE-M3, and $0 \leq w_b \leq 1$. Adding E5 gives

$$S_{bge}(q, d) = w_b \widetilde{s}_b(q, d) + w_g \widetilde{s}_g(q, d) + w_e \widetilde{s}_e(q, d),$$
$$w_b + w_g + w_e = 1,$$

where e denotes E5 and all weights are nonnegative. The articles are ranked by the combined score in decreasing order.

The baseline selects weights from a grid with increments of 0.1. BM25+BGE-M3 has 11 candidate settings, with $w_b \in \{0, 0.1, \ldots, 1\}$. BM25+BGE-M3+E5 has 66 candidate settings whose weights sum to one. This grid includes zero-weight settings, including $w_e = 0$, so it can reproduce every BM25+BGE-M3 setting in the baseline grid. A higher score on the weight-selection queries therefore does not establish that E5 improves retrieval on separate evaluation queries.

For each baseline split, the weights with the highest mean NDCG@10 on the weight-selection queries are fixed before evaluation on the other queries. Section 4.8 describes the grouping and fold procedure. Algorithm 4.1 shows how the selected configuration produces a ranking for one query.

Algorithm 4.1 Article ranking with fixed fusion weights

Require: Query q, corpus C, retriever scores, fixed fusion weights w
Ensure: Ten highest-ranked articles
1: Normalize each retriever's scores over the full corpus for q
2: **for** each article $d \in C$ **do**
3: combined$[d] \leftarrow \sum_r w_r$ normalized_score$[r, d]$
4: **end for**
5: **return** the ten articles with highest combined score

The fine-tuning comparisons use weights selected with pretrained BGE-M3 and keep them fixed when replacing it with the fine-tuned model. BM25 and E5 also remain unchanged. These comparisons measure the effect of replacing BGE-M3 within the specified hybrid configuration. The 0.1 grid described above applies to baseline selection. The fine-tuning experiments use separate selection procedures, described in Section 4.7.

4.7 BGE-M3 Fine-Tuning

The two fine-tuning experiments differ in how they select BGE-M3: the first uses hybrid retrieval performance, while the second uses standalone dense retrieval performance. Both then compare hybrids containing pretrained and fine-tuned BGE-M3. BM25, E5, and the selected combination weights remain fixed within each pretrained and fine-tuned comparison. The experiments use different training procedures and are therefore described separately.

4.7.1 Model Selection Based on Hybrid Retrieval

Training Data and Negative Examples

This experiment uses Zalo question and article pairs. For each eligible query, the training builder selects a known relevant article available in both clean and parsed views, creating paired training examples from the two corpus representations. Negative passages are mined from lexical and dense rankings after excluding known positives, duplicates, unavailable documents, and laws reserved for development and test evaluation.

The negative-selection procedure considers up to 256 candidates from each of BM25 and pretrained BGE-M3 on the parsed corpus. It alternates between the filtered lists until seven distinct negatives have been selected. These rankings and negative groups are fixed before training. All explicit negatives use parsed article text, including groups whose positive uses clean text.

Table 4.4: Training groups and protected folds in the completed hybrid fine-tuning study.

Stage	Train folds	Queries	A/B groups	Protected folds	Protected laws	Law overlap
Pilot	2, 3, 4	1,917	3,834	0, 1	258	0
Final training	1, 2, 3, 4	2,556	5,112	0	110	0

Each eligible query produces two training groups using the clean and parsed text of the same selected relevant article. The group counts therefore do not represent independent questions. The zero-overlap checks confirm that training groups contain no article from the protected law identifiers; they do not establish separation of legal concepts or similar

language. Explicit-negative filtering also does not guarantee that passages from other groups in the same batch are nonrelevant to a query.

Training Objective and Configuration

This experiment trains BGE-M3 with a single-positive in-batch contrastive objective. Each query group contains one selected positive and seven explicit negatives. With batch size B, group size $G = 8$, positive passage index $y_i = iG$, normalized query vector $\mathbf{q}_i$, normalized passage vector $\mathbf{p}_j$, and temperature $\tau = 0.02$, the implemented objective is

$$\mathcal{L}_{\text{batch}} = -\frac{1}{B}\sum_{i=0}^{B-1}\log\frac{\exp(\mathbf{q}_i^\top \mathbf{p}_{y_i}/\tau)}{\sum_{j=0}^{BG-1}\exp(\mathbf{q}_i^\top \mathbf{p}_j/\tau)}.$$

Table 4.5: Locked configuration for the completed final-training runs.

Parameter	**Verified value**
Base encoder	BGE-M3
Trainable component	Full dense encoder
Epochs	2
Learning rate	2.0000×10^{-5}
Contrastive query batch	4
Gradient accumulation	8
Optimizer effective batch	32 query groups
Passage group	1 selected positive and 7 explicit negatives
Query / passage maximum length	256 / 512 tokens
Temperature	0.0200
Pooling / normalization	First-token hidden state / L2 normalization
Seeds	42, 43, 44
Frozen E5 component	Multilingual-E5-base, unchanged throughout fine-tuning

The numerator rewards the selected positive article, while the denominator includes all passages in the forward-pass batch. With four query groups and eight passages per group, this denominator contains 32 passage positions. Gradient accumulation across eight forward passes yields an effective optimizer batch of 32 query groups, but does not combine their passages into one contrastive denominator.

Model Selection and Evaluation

Pilot training uses folds 2, 3, and 4. Fold 1 selects the learning rate, epoch, and forward-pass query batch using hybrid NDCG@10. NDCG@10 for BM25+BGE-M3 is the primary selection criterion, while BM25+BGE-M3+E5 is checked separately. The development rule requires an improvement of at least 0.01 for BM25+BGE-M3 and a decrease of no more than 0.01 for BM25+BGE-M3+E5. After the configuration is fixed, final models are trained again from pretrained BGE-M3 on folds 1, 2, 3, and 4 with seeds 42, 43, and 44.

Fold 0 contains 640 queries reserved for evaluation on a separate test set. The corrected comparison selects combination weights on folds 1, 2, 3, and 4 using pretrained BGE-M3 and keeps them fixed when replacing it with fine-tuned BGE-M3. A preliminary analysis had already examined fold 0 before a document-identity error was corrected. The corrected results are therefore interpreted as post-hoc evidence, despite excluding fold 0 from training and model selection.

Figure 4.3 summarizes the selection and evaluation sequence.

4.7.2 Model Selection Based on Standalone Dense Retrieval

Training Data and Configuration

This experiment selects BGE-M3 using standalone dense retrieval before testing it in BM25+BGE-M3 and BM25+BGE-M3+E5. It uses a fixed corpus of 61,425 parsed articles, with reference text retained where parsed text is unavailable. The official training partition contains 2,556 queries. Removing one near-duplicate of an official test query leaves 2,555 eligible training groups. The final evaluation uses 640 official test queries.

Each group includes five explicit negatives. The training procedure uses a mask to identify all known positive articles within the batch candidate set. This differs from the selected-positive objective in Section 4.7.1. The encoder uses the hidden state at the first token position, L2 normalization, and query and passage limits of 256 and 512 tokens, respectively. The specified temperature is 0.05, with bfloat16 precision, a cosine learning-rate schedule, and a warmup ratio of 0.05. The procedure requires at least 32 queries in a forward-pass contrastive batch, with 64 preferred; this requirement cannot be met by gradient accumulation alone. The available description does not identify which of these batch sizes was used in every final run.

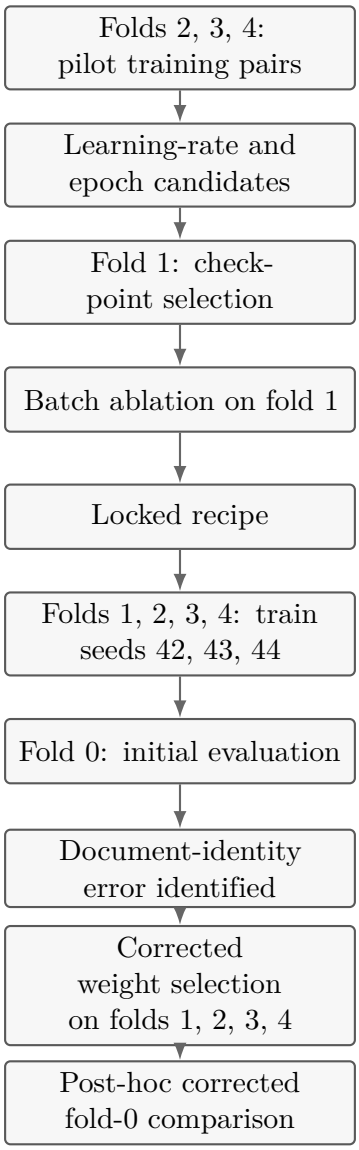


Figure 4.3: Training and evaluation evidence flow.

Training Objective

For logits $z_{ij} = s(q_i, d_j)/\tau$, positive candidate positions P_i, candidate positions C_i, and batch size B, the multi-positive loss is

$$\mathcal{L} = -\frac{1}{B}\sum_{i=1}^{B}\left[\log\sum_{j\in P_i}\exp(z_{ij}) - \log\sum_{j\in C_i}\exp(z_{ij})\right].$$

Here $s(q_i, d_j)$ is the similarity between query i and candidate article j, and $\tau = 0.05$ controls the scale of the logits. The numerator includes every known positive present in the candidate set rather than one selected positive. This use of recorded labels does not establish that every unannotated candidate is legally irrelevant. The documented checks concern article identity, labels, and data provenance; they do not constitute expert review of every negative example.

Model Selection and Hybrid Evaluation

Screening evaluates 36 checkpoints across three learning rates, four development folds, and training steps 30, 60, and 90. A training-data stage passes if its mean NDCG@10 gain is at least 0.01, at least three folds improve NDCG@10, and mean Recall@100 decreases by no more than 0.01. These criteria use standalone dense retrieval. The Zalo-only stage is the first stage to pass, so subsequent data expansions are skipped. Screening selects a learning rate of 2×10^{-5} and step 60. Final models are trained using seeds 42, 43, and 44, with each checkpoint taken at step 60 and checked after reloading.

For the hybrid comparisons, combination weights are selected on official training queries using pretrained BGE-M3, then fixed when fine-tuned BGE-M3 replaces it. BM25+BGE-M3 uses BM25 and BGE-M3 weights of 0.20 and 0.80. BM25+BGE-M3+E5 uses BM25, BGE-M3, and E5 weights of 0.15, 0.50, and 0.35. Dense and hybrid comparisons use the same 640 test queries. A gain of at least 0.01 NDCG@10 defines the practical improvement threshold, with the Recall@100 condition checked separately. This threshold is a decision rule rather than a statistical significance test.

The official test queries had been examined before this experiment. Excluding them from final training and model selection does not remove that prior exposure, so the results are interpreted as post-hoc evidence. They assess query retrieval over a fixed corpus and do not establish transfer to new legal documents or a later time period. Chapter 5 reports standalone and hybrid outcomes separately.

4.8 Evaluation Design and Statistical Analysis

4.8.1 Query Grouping and Data Splits

Queries that cite the same legal document are linked into groups. These links are transitive: if one question cites documents shared with two other questions, all three belong to the same connected group. Keeping each group within one fold prevents a cited document from appearing on both sides of fusion-weight selection and evaluation. It does not eliminate similarity in legal concepts or wording across groups.

For the pretrained ALQAC and Zalo comparisons, fusion weights are selected on the other folds and evaluated on the excluded fold. Each query receives one out-of-fold result, meaning that its relevance targets are excluded from the weight selection used for its prediction. The encoders remain fixed during this procedure. The ALQAC query pool combines the original training and private-test gold files, so these results describe grouped evaluation over that combined pool rather than performance on the original private-test partition.

Algorithm 4.2 specifies the selection procedure. The overall score is calculated from all evaluation queries, giving each query equal weight. It is not an unweighted mean of fold scores when fold sizes differ. Because each fold can select different weights, the result evaluates the selection procedure rather than one fixed weight vector.

Algorithm 4.2 Grouped selection of fusion weights and out-of-fold evaluation

Require: Ordered queries Q, corpus D, aligned normalized score matrices, grouped folds, ordered candidate weight grid W
Ensure: Selected weights per fold and evaluation metrics for every query
for each fold f **do**
  Set T to the selection-query indices and V to the evaluation-query indices
  for each candidate weight vector $w \in W$, in recorded order **do**
    Rank D for every query in T using the weighted score sum
    Compute mean query NDCG@10
  end for
  Select the first candidate with the highest selection mean
  Rank all queries in V using the selected weights
  Store each evaluation query's metrics
end for
Aggregate evaluation-query rows across folds, giving each query equal weight

BCA requires a separate allocation because 484 of its 546 queries form one connected group. The remaining 62 queries belong to 49 smaller groups. The first comparison selects weights using the 62 queries and evaluates on the 484-query group. The smaller groups are then assigned to four evaluation folds containing 16, 16, 15, and 15 queries. For each of these folds, weights are selected on the remaining queries, including the largest group. Each question receives one evaluation result, but the selection populations differ substantially. The combined BCA score is therefore a descriptive summary, not a balanced

five-fold estimate.

The fine-tuning experiments use the separate training, development, and test allocations described in Section 4.7. Their results are compared with pretrained models on the same test queries, not with the all-fold pretrained averages. Although test queries are excluded from the corresponding training and model selection, they had been examined in earlier analyses. Both fine-tuning comparisons are therefore interpreted as post-hoc evidence.

4.8.2 Evaluation Metrics and Comparisons

The target-matching rules and metric formulas are defined in Section 3.6. NDCG@10 is the primary metric. Hit@10 measures whether at least one target is recovered, MRR@10 measures the position of the first match, and MAP@10 summarizes precision at matching ranks. The standalone dense and corresponding hybrid fine-tuning comparisons also report Recall@100.

The comparison of BM25+BGE-M3 with BM25 measures improvement over the lexical baseline. The comparison of BM25+BGE-M3+E5 with BM25+BGE-M3 isolates the contribution of adding E5 under the specified selection procedure. Fine-tuning comparisons replace pretrained BGE-M3 with its fine-tuned version while holding BM25, E5, and the corresponding fusion weights fixed. Standalone dense performance is assessed separately from hybrid performance. The specified threshold for a practically useful hybrid improvement is an absolute increase of 0.01 NDCG@10; this threshold is distinct from a statistical significance criterion.

BCA results are reported both for all retained queries and for queries with at least one target available in the corpus. The latter population still includes queries with only some targets available, and their original metric denominators are retained. This distinction separates the effect of target availability from ranking performance without treating incomplete annotations as complete relevance judgments.

4.8.3 Statistical Analysis

For ALQAC and Zalo, the recorded comparison between BM25+BGE-M3+E5 and BM25+BGE-M3 uses the Wilcoxon signed-rank test on paired query-level out-of-fold NDCG@10 and MRR@10 values. Its `zero_method=zsplit` setting splits the ranks of zero differences between the positive and negative rank sums. These stored tests are unadjusted for the multiple condition and metric comparisons. Queries within the same legal group can remain dependent despite grouped weight selection, so the query-level tests do not establish variation across independent legal topics.

For BCA's 49 smaller groups, the analysis first averages paired NDCG@10 differences

within each group. It then applies 100,000 random sign flips to those group means and uses 10,000 bootstrap resamples of the group means to form a 95% interval. Each group has equal weight in these procedures. Their interval therefore concerns the mean across groups, whereas the reported query-level mean gives each question equal weight. These quantities differ when group sizes differ. Query resampling within the single largest group describes variation among its questions and cannot estimate variation across multiple such legal groups. The combined BCA aggregate remains descriptive.

For the corrected fine-tuning comparison selected by hybrid performance, fine-tuned NDCG@10 is first averaged across seeds 42, 43, and 44 separately for each query. The matched pretrained value is then subtracted. The analysis uses 2,000 seeded bootstrap resamples of 106 legal groups covering the 640 test queries. Each resample draws groups with replacement, includes all their query differences, and calculates the query mean. Percentiles of these means form the reported 95% interval.

That implementation also records twice the smaller proportion of bootstrap means on either side of zero, capped at one, followed by Holm adjustment across the two hybrid comparisons. This is a bootstrap tail-area quantity, not a Wilcoxon result. Because the resamples come from the observed differences rather than an explicitly generated null distribution, it is not treated as a calibrated hypothesis-test p-value. Unequal group sizes also affect the resampling distribution: one legal group contains 428 of the 640 queries.

Averaging within each query avoids counting three seed-specific predictions as three independent questions. The resulting interval concerns the average of the three observed seed results; it does not estimate uncertainty over arbitrary future seeds or evaluate an ensemble of encoders. For both fine-tuning experiments, variation across the three seeds is reported separately from uncertainty over queries. An interval containing zero does not establish equivalence, and a numerical improvement does not by itself establish a statistically supported or practically useful gain.

4.9 Experimental Environment and Reproducibility

The recorded execution environments differ between the two fine-tuning experiments. Both report an NVIDIA L4 graphics processing unit and PyTorch `2.11.0+cu128`. The experiment using hybrid retrieval for model selection records Transformers `5.15.0`, NumPy `2.1.3`, and 16-bit floating-point (FP16) precision. The experiment using standalone dense retrieval for model selection records Transformers `5.16.1` and bfloat16 (BF16) precision. These details describe the recorded training runs rather than the local computer used to prepare the thesis. The final hybrid-selected run records do not specify processor, system memory, graphics memory, or driver details; preliminary environment checks cannot establish those details for the completed runs.

Each selected fine-tuning configuration is evaluated with training seeds 42, 43, and 44. Reproducing the comparison requires the same model revision and tokenizer, training examples, query and article ordering, data splits, checkpoint selection rule, and fusion coefficients. Appendix D records the hybrid-selected configuration, model revision, and checksums for its model files, training data, and training code. These records identify the inputs used in the experiment; access to the checkpoint files is also required to reproduce encoder outputs.

The saved score matrices and selected fusion coefficients support recalculation of the reported retrieval metrics without retraining. Their rows and columns must remain aligned with the recorded query and article identifiers. Recalculating metrics verifies the numerical results but does not establish that model training can be repeated exactly. Appendix F describes the reproduction procedure and its limits.

The experiments assess retrieval effectiveness. They do not provide controlled measurements of query latency, throughput, or deployment cost, so the recorded hardware specifications do not support conclusions about those outcomes.

4.10 Research Ethics and Data Use

The experiments use the ALQAC and Zalo collections and the BCA dataset described in Sections 4.2 and 4.3. BCA is derived from public questions and published answers on the Bộ Công an portal. Public access identifies the source of these records, but does not by itself establish permission for unrestricted reuse or redistribution. The construction records described in this thesis do not establish consent from the people who submitted the questions or document a completed review of personal information.

Public questions and answers may contain details that identify individuals. Any future release of BCA therefore requires a review of identifying information, source terms, and the permitted scope of redistribution. The example in Section 4.3 summarizes the query and target structure without reproducing personal details. Published answers are retained as evidence for reference extraction and are not supplied as query inputs during retrieval.

The evaluation measures recovery of recorded legal references. These references are extracted from published answers and may be incomplete or incorrectly mapped, as discussed in Section 4.3. Retrieval scores therefore do not establish the correctness of a legal answer or the current applicability of a retrieved provision. The experiments assess article ranking and do not evaluate a system for providing legal advice.

4.11 Chapter Summary

This chapter presents the research design, the datasets and corpus preparation, and the construction of BCA from public questions and published answers. It specifies lexical and dense retrieval, score normalization and fusion, and BGE-M3 fine-tuning with model selection based on either hybrid or standalone dense retrieval. The evaluation design explains query grouping, data splits, metric comparisons, and the scope of statistical analysis. The chapter also documents the experimental environment, reproducibility requirements, and data-use limitations; Chapter 5 reports the results.

Chapter 5

Experimental Evaluation

This chapter reports the experiments in the order of the research questions. It first evaluates lexical and hybrid retrieval, then isolates the incremental contribution of E5, then examines BCA coverage and component-aware reliability, and finally analyzes legal-domain adaptation. The central adaptation finding is that dense retriever improvement does not necessarily transfer into a complete hybrid retrieval system.

5.1 Datasets and Experimental Setup

Quantitative table values are displayed to four decimal places, using scientific notation with four fractional digits in the mantissa for tiny nonzero values. Deltas are calculated before display rounding; integer counts and indices remain integers.

The evaluated conditions and corpus sizes are defined in Table 4.1. ALQAC and Zalo each have clean and parsed corpus representations; their A/B conditions share query identities and are therefore paired corpus conditions rather than independent datasets. BCA is evaluated separately because many questions share legal targets and because some targets are unavailable in the indexed corpus.

NDCG@10 is the primary endpoint. Hit@10, MRR@10, and MAP@10 are reported to distinguish top-ten recovery, first-relevant ordering, and multiple-target ranking. ALQAC and Zalo use query-level paired comparisons for the two-way versus three-way hybrid. BCA uses component-aware analysis in addition to descriptive query-level summaries. These units of inference are not interchangeable.

Table 5.1: System identities and comparison boundaries.

Configuration	**Baseline role**	**Adaptation role**
BM25	Fixed lexical reference	Unchanged lexical component
BGE-M3 dense	Pretrained dense retriever	Adapted in the legal-domain fine-tuning studies
E5 dense	Additional pretrained dense retriever	Frozen throughout the thesis

Continued on next page

Configuration	Baseline role	Adaptation role
BM25+BGE-M3	Two-way lexical–semantic hybrid	Evaluated with pretrained and adapted BGE-M3
BM25+ BGE-M3+E5	Three-way hybrid	Evaluated with pretrained and adapted BGE-M3 while E5 remains fixed

The experiments compare methods within their own query populations and selection boundaries. Results from different folds, corpora, or adaptation recipes are not pooled into a single cross-protocol score.

5.2 Lexical and Hybrid Retrieval

This section compares lexical–semantic hybrid retrieval with the fixed BM25 reference. Table 5.2 reports BM25, pretrained dense retrievers, and hybrid systems for ALQAC and Zalo. Table 5.3 summarizes the hybrid changes relative to BM25 and includes BCA's combined descriptive view.

Table 5.2: Pretrained standalone and hybrid results, query-macro OOF population.

Condition	Method	NDCG@10	MRR@10	MAP@10	Hit@10
ALQAC A	BM25	0.7986	0.7648	0.7621	0.9112
ALQAC A	BGE-M3 pretrained	0.8437	0.8134	0.8108	0.9433
ALQAC A	E5 pretrained	0.8126	0.7729	0.7722	0.9371
ALQAC A	BM25+BGE-M3	0.8805	0.8534	0.8516	0.9692
ALQAC A	BM25+BGE-M3+ E5	0.8784	0.8519	0.8504	0.9630
ALQAC B	BM25	0.6479	0.6175	0.6159	0.7472
ALQAC B	BGE-M3 pretrained	0.6684	0.6401	0.6382	0.7620
ALQAC B	E5 pretrained	0.6510	0.6177	0.6165	0.7583
ALQAC B	BM25+BGE-M3	0.7067	0.6812	0.6797	0.7904
ALQAC B	BM25+BGE-M3+ E5	0.7055	0.6805	0.6793	0.7867
Zalo A	BM25	0.5264	0.4731	0.4682	0.7100
Zalo A	BGE-M3 pretrained	0.6424	0.5917	0.5850	0.8235

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Condition	Method	NDCG@10	MRR@10	MAP@10	Hit@10
Zalo A	E5 pretrained	0.5640	0.5104	0.5046	0.7509
Zalo A	BM25+BGE-M3	0.6805	0.6298	0.6235	0.8583
Zalo A	BM25+BGE-M3+ E5	0.6818	0.6317	0.6254	0.8576
Zalo B	BM25	0.5620	0.5099	0.5050	0.7406
Zalo B	BGE-M3 pretrained	0.6889	0.6431	0.6366	0.8526
Zalo B	E5 pretrained	0.6259	0.5748	0.5689	0.8054
Zalo B	BM25+BGE-M3	0.7214	0.6767	0.6705	0.8789
Zalo B	BM25+BGE-M3+ E5	0.7344	0.6913	0.6845	0.8889

BGE-M3 is the strongest individual retriever by NDCG@10 in all four ALQAC/Zalo conditions. Both hybrids exceed BM25, and both also exceed the strongest individual component in these conditions. The result supports lexical–semantic complementarity at the system level, although it does not identify the linguistic mechanism behind each gain.

Table 5.3: Hybrid NDCG@10 changes relative to fixed BM25.

Condition	Hybrid	BM25	Hybrid score	Absolute change	Relative change
ALQAC A	BM25+ BGE-M3	0.7986	0.8805	+0.0819	+10.2590%
ALQAC A	BM25+ BGE-M3+E5	0.7986	0.8784	+0.0798	+9.9970%
ALQAC B	BM25+ BGE-M3	0.6479	0.7067	+0.0588	+9.0690%
ALQAC B	BM25+ BGE-M3+E5	0.6479	0.7055	+0.0576	+8.8910%
Zalo A	BM25+ BGE-M3	0.5264	0.6805	+0.1541	+29.2760%
Zalo A	BM25+ BGE-M3+E5	0.5264	0.6818	+0.1554	+29.5210%
Zalo B	BM25+ BGE-M3	0.5620	0.7214	+0.1594	+28.3680%
Zalo B	BM25+ BGE-M3+E5	0.5620	0.7344	+0.1724	+30.6830%
BCA combined descriptive	BM25+ BGE-M3	0.1687	0.2283	+0.0597	+35.3820%

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Condition	Hybrid	BM25	Hybrid score	Absolute change	Relative change
BCA combined descriptive	BM25+ BGE-M3+E5	0.1687	0.2297	+0.0610	+36.1760%

The hybrid gains over BM25 are large in Zalo and smaller but still positive in ALQAC. BCA also shows a positive descriptive difference from BM25, but its lower absolute scores must be interpreted with coverage limits and component dependence. The answer to RQ1 is therefore positive within the evaluated conditions: lexical–semantic hybrids improve the fixed BM25 reference. The result should not be generalized into a claim that one hybrid configuration will transfer unchanged to every legal corpus.

5.3 Incremental Contribution of E5

This section tests whether E5 adds value beyond BM25+BGE-M3. This is a stricter question than asking whether E5 is a competent standalone retriever. The relevant comparison is the two-way hybrid against the three-way hybrid.

Table 5.4: Incremental E5 contribution on the canonical ALQAC and Zalo populations.

Condition	Two-way NDCG	Three-way NDCG	Absolute change	Relative change	Stored Wilcoxon p
ALQAC A	0.8805	0.8784	-0.0021	-0.2380%	0.4498
ALQAC B	0.7067	0.7055	-0.0012	-0.1630%	0.7449
Zalo A	0.6805	0.6818	+0.0013	+0.1900%	0.1548
Zalo B	0.7214	0.7344	+0.0130	+1.8030%	5.1598×10^{-7}

The incremental E5 result is condition-dependent. ALQAC shows small negative NDCG@10 differences in both corpus representations. Zalo A shows a very small positive difference, while Zalo B shows the clearest positive result: an absolute NDCG@10 gain of approximately 0.0130, with higher MRR@10 and Hit@10 as well. The p-values in Table 5.4 are reported as stored query-level paired tests; they do not remove the need to interpret multiple conditions and possible query dependence carefully.

The A/B corpus comparison also resists a simple parser narrative. ALQAC has substantially lower hybrid scores in the parsed condition than in the clean condition. Zalo, in contrast, has higher scores in the parsed-with-fallback condition than in the clean condition. This does not imply that parsing is inherently harmful for ALQAC and inherently beneficial for Zalo. The two collections differ in construction, target mapping, fallback behavior, and

article universe. A matched no-fallback ablation was not available, so the independent effect of fallback remains unmeasured.

The answer to RQ2 is therefore not a general yes. E5 provides useful incremental value in the parsed Zalo condition, negligible or unsupported value in clean Zalo, and no observed improvement in ALQAC. This is one of the thesis's main practical findings: adding a dense retriever expands the search space, but it does not guarantee better ranking on unseen data.

5.4 BCA Coverage and Component-Aware Evaluation

BCA serves two purposes. It extends the evaluation beyond ALQAC and Zalo, and it exposes how corpus coverage and target dependence affect interpretation. Table 5.5 reports BCA results under the giant-component, small-component, and combined descriptive views.

Table 5.5: Compact BCA protocol and coverage views. All metrics are at cutoff 10; OOF denotes out-of-fold.

Protocol	**Cohort**	**Method**	**N**	**NDCG**	**MRR**	**MAP**	**Hit**
Giant	All	BM25	484	0.1670	0.2469	0.1098	0.4545
Giant	All	BGE-M3	484	0.2153	0.3143	0.1453	0.5455
Giant	All	BM25+BGE	484	0.2275	0.3414	0.1573	0.5517
Giant	All	BM25+BGE+E5	484	0.2295	0.3413	0.1560	0.5702
Giant	Answerable	BM25	394	0.2051	0.3033	0.1348	0.5584
Giant	Answerable	BGE-M3	394	0.2645	0.3861	0.1785	0.6701
Giant	Answerable	BM25+BGE	394	0.2795	0.4194	0.1932	0.6777
Giant	Answerable	BM25+BGE+E5	394	0.2819	0.4193	0.1916	0.7005
Small OOF	All	BM25	62	0.1817	0.2157	0.1382	0.3387
Small OOF	All	BGE-M3	62	0.2015	0.2392	0.1597	0.3387
Small OOF	All	BM25+BGE	62	0.2346	0.2921	0.1865	0.3871
Small OOF	All	BM25+BGE+E5	62	0.2309	0.2838	0.1815	0.3871
Small OOF	Answerable	BM25	46	0.2449	0.2907	0.1863	0.4565
Small OOF	Answerable	BGE-M3	46	0.2715	0.3225	0.2153	0.4565
Small OOF	Answerable	BM25+BGE	46	0.3162	0.3937	0.2514	0.5217
Small OOF	Answerable	BM25+BGE+E5	46	0.3112	0.3825	0.2446	0.5217
Combined OOF	All	BM25	546	0.1687	0.2434	0.1130	0.4414
Combined OOF	All	BGE-M3	546	0.2137	0.3058	0.1469	0.5220
Combined OOF	All	BM25+BGE	546	0.2283	0.3358	0.1606	0.5330
Combined OOF	All	BM25+BGE+E5	546	0.2297	0.3348	0.1589	0.5495
Combined OOF	Answerable	BM25	440	0.2093	0.3020	0.1402	0.5477
Combined OOF	Answerable	BGE-M3	440	0.2652	0.3795	0.1823	0.6477
Combined OOF	Answerable	BM25+BGE	440	0.2833	0.4167	0.1993	0.6614
Combined OOF	Answerable	BM25+BGE+E5	440	0.2850	0.4154	0.1972	0.6818

The BCA reference graph contains one giant component with 484 of 546 queries and 49 smaller components containing the remaining 62 queries. This structure affects statistical

interpretation. The combined descriptive view shows the three-way hybrid slightly above the two-way hybrid in NDCG@10, but the small-component out-of-fold view shows a negative NDCG@10 difference for the three-way system. The corresponding component-aware inference does not establish an E5 advantage.

Coverage is also a major limitation. Of the 546 BCA queries, 440 have at least one represented target and 106 have none. A further 229 answerable queries are partial-target cases in which some, but not all, targets are present. An unanswerable query cannot receive a positive Hit@10 from any ranking method over the fixed corpus. Therefore, BCA's lower end-to-end scores partly reflect corpus and mapping limitations rather than only scoring failure.

The BCA result is a dataset and evaluation contribution rather than a simple model leaderboard. It shows that an end-to-end legal retrieval score can mix three phenomena: whether the evidence exists, whether it is mapped to the correct legal identity, and whether the retriever ranks it highly. A method improvement can address only the third problem unless corpus construction changes as well.

5.5 Failure Analysis

The failure analysis asks what a model intervention could plausibly repair. Missing targets require corpus or mapping work. Available targets outside the top ten require ranking or fusion improvements. Relevant articles demoted within the top ten affect NDCG and MRR even when Hit@10 is unchanged. These distinctions guide the adaptation experiment but do not prove that fine-tuning will solve the observed failures.

BCA provides the clearest evidence of non-rankable failures: 106 questions have no mapped target in the corpus. For these questions, changing BM25 weights, dense encoders, or fusion rules cannot recover the target. Partial-target queries create a second limitation: a system may retrieve an available relevant article while still being penalized for missing other targets that are absent from the corpus.

For ALQAC and Zalo, top-ten boundary cases show that E5 can both recover and lose hits. The same added signal that improves parsed Zalo on average can demote specific relevant articles below the cutoff. Appendix E preserves representative cases. The main conclusion is more general: aggregate improvement does not imply uniform query-level improvement, and aggregate regression does not imply that every query becomes worse.

The analysis remains an information-retrieval analysis. It identifies target absence, top-ten transitions, within-law article demotion, and persistent misses. It does not assign legal-reasoning causes such as misunderstanding exceptions or temporal validity. A legally adjudicated error taxonomy would require a sampling plan, full text review, annotation rules, and reviewer agreement. That study was not performed here.

5.6 Legal-Domain Adaptation

This section examines whether legal-domain BGE-M3 adaptation improves retrieval beyond development selection. The narrative has six parts: Study A development selection, Study A corrected hybrid evaluation on unseen data, Study B dense-only adaptation, transfer of dense improvements into hybrid retrieval, query-level analysis, and overall interpretation.

5.6.1 Development Selection

Study A: Hybrid-Oriented Adaptation selects a BGE-M3 training recipe on development queries. Table 5.6 reports the learning-rate and epoch pilot under passage length 512. Table 5.7 reports the subsequent batch ablation at the selected learning rate and epoch.

Table 5.6: Passage-512 learning-rate and epoch pilot: development NDCG@10.

Learning rate	**Epoch**	**Two-way hybrid**	**Three-way hybrid**
5.0000×10^{-6}	1	0.7470	0.7685
5.0000×10^{-6}	2	0.7603	0.7785
5.0000×10^{-6}	3	0.7647	0.7815
1.0000×10^{-5}	1	0.7509	0.7816
1.0000×10^{-5}	2	0.7658	0.7873
1.0000×10^{-5}	3	0.7628	0.7897
2.0000×10^{-5}	1	0.7440	0.7757
2.0000×10^{-5}	2	0.7691	0.7923
2.0000×10^{-5}	3	0.7689	0.7958

The pilot shows development improvements across several candidate settings. The selected recipe uses learning rate 0.00002 and two epochs. The choice is not evidence that the selected point is a global optimum; it is the best supported point under the restricted search and selection rule.

Table 5.7: Batch ablation at the selected learning rate and epoch; scores are development NDCG@10.

Batch/accum.	**Two-way**	**Δ**	**Three-way**	**Δ**
Pretrained baseline	0.7172	–	0.7245	–
1 / 32	0.7725	+0.0553	0.7869	+0.0624
2 / 16	0.7693	+0.0521	0.7924	+0.0679
4 / 8	0.7781	+0.0609	0.7934	+0.0689

The selected development configuration improves two-way NDCG@10 by 0.060945 and three-way NDCG@10 by 0.068882. This justifies testing the intervention on unseen data. It does not establish generalization, because the development queries are part of model selection.

5.6.2 Corrected Hybrid Evaluation on Unseen Data

The corrected comparison on unseen data evaluates Zalo parsed fold 0 with 640 queries. Fusion weights are fitted on the permitted training-side folds using pretrained BGE-M3 and then held fixed when BGE-M3 is replaced by its adapted version. This design measures encoder replacement within a specified hybrid rather than unrestricted retuning of the whole system.

The corrected two-way weights are BM25/BGE/E5 = 0.2/0.8/0.0. The corrected three-way weights are 0.2/0.5/0.3. Because a flawed preliminary analysis had already exposed the same fold, the corrected comparison is interpreted as a post-hoc evaluation on data unseen during training and selection, not as a pristine first-use test.

Table 5.8: Corrected Zalo fold-0 two-way hybrid evaluation; all metrics are at cutoff 10.

BGE state	Seed	NDCG	Δ	MRR	MAP	Hit
Pretrained	Fixed	0.7516	–	0.7015	0.6985	0.9141
Fine-tuned	42	0.7005	−0.0511	0.6470	0.6451	0.8750
Fine-tuned	43	0.7005	−0.0511	0.6490	0.6467	0.8688
Fine-tuned	44	0.6977	−0.0539	0.6398	0.6374	0.8859
Fine-tuned	Mean	0.6995	−0.0520	0.6453	0.6430	0.8766

All three two-way seeds score below the pretrained two-way baseline. The mean NDCG@10 change is -0.052045, and MRR@10, MAP@10, and Hit@10 are also lower. The result therefore cannot be rescued by choosing a different seed from the three reported runs.

Table 5.9: Corrected Zalo fold-0 three-way hybrid evaluation; all metrics are at cutoff 10.

BGE state	Seed	NDCG	Δ	MRR	MAP	Hit
Pretrained	Fixed	0.7595	–	0.7077	0.7048	0.9266
Fine-tuned	42	0.7320	−0.0275	0.6771	0.6743	0.9141
Fine-tuned	43	0.7340	−0.0255	0.6771	0.6747	0.9203
Fine-tuned	44	0.7378	−0.0217	0.6794	0.6771	0.9281
Fine-tuned	Mean	0.7346	−0.0249	0.6779	0.6754	0.9208

The three-way hybrid also has lower mean NDCG@10 after fine-tuning. The regression is smaller than in the two-way system, but it remains a regression against the matched pretrained three-way comparator. Seed 44 slightly improves Hit@10 while reducing

NDCG@10 and MRR@10, showing why the primary rank-sensitive metric must remain central.

Table 5.10: Corrected paired statistics for NDCG@10. Bootstrap tail values are calculated as described in Section 4.8; they are not calibrated hypothesis-test p-values.

Comparison	Queries	Groups	Mean Δ	95% group bootstrap interval	Bootstrap tail value	Holm-adjusted tail value
Fine-tuned minus pre-trained, BM25+BGE-M3	640	106	−0.0520	[−0.0919, +0.1076]	0.7480	1.0000
Fine-tuned minus pre-trained, BM25+BGE-M3+E5	640	106	−0.0249	[−0.0542, +0.0938]	0.7480	1.0000

The intervals in Table 5.10 include zero, so the observed regressions are not described as statistically established population-level harm. However, non-significance is not equivalence, and the practical improvement threshold of +0.01 NDCG@10 is not met. The correct conclusion is that the selected adaptation recipe failed to demonstrate hybrid improvement on unseen data under this protocol.

5.6.3 Study B: Dense-Oriented Adaptation

Study A does not by itself determine whether adaptation failed at the dense-retriever level or only after integration into the hybrid. Study B directly evaluates dense-only BGE-M3 adaptation before testing hybrid transfer.

Table 5.11: Study B post-hoc dense benchmark on 640 official queries.

System	NDCG@10	Delta NDCG@10	MRR@10	Recall@100
Pretrained	0.6740	0.0000	0.6213	0.9617
Fine-tuned seed 42	0.6932	0.0192	0.6448	0.9531
Fine-tuned seed 43	0.6925	0.0185	0.6431	0.9516
Fine-tuned seed 44	0.6928	0.0189	0.6437	0.9531
Fine-tuned seed mean	0.6928	0.0189	0.6438	0.9526

Study B shows a mean NDCG@10 gain of 0.018866, with all three seeds improving this endpoint. Mean Recall@100 decreases by 0.009115. The dense result therefore supports the claim that legal-domain fine-tuning can improve the dense encoder on this fixed-corpus query benchmark, while also showing a small recall tradeoff at depth 100.

5.6.4 Transfer of Dense Improvements into Hybrid Retrieval

The crucial question is whether the dense improvement remains useful inside the full hybrid. Table 5.12 reports the Study B primary locked-weight hybrid outcomes. Fusion weights are selected with pretrained BGE-M3 and then held fixed for the corresponding fine-tuned comparison.

Table 5.12: Study B primary fusion outcomes under pretrained-locked weights.

Hybrid	Pretrained	FT mean	Δ NDCG	Rel. Δ	Δ R@100	Positive seeds	Gate
Two-way	0.7526	0.7483	-0.0043	-0.5707%	-0.0057	0/3	Fail
Three-way	0.7467	0.7476	0.0009	0.1203%	-0.0031	2/3	Fail

The dense-only gain does not transfer into a practically useful hybrid gain. The two-way hybrid decreases by 0.004295 NDCG@10. The three-way hybrid increases by only 0.000898 NDCG@10, far below the +0.01 improvement threshold. Thus, Study B sharpens the main RQ3 finding: adaptation can improve standalone dense ranking while failing to improve the complete hybrid retrieval system by a meaningful amount.

The mechanism is not isolated by these experiments. Possible explanations include changed score calibration, overlap between lexical and dense evidence, and fusion weights selected for pretrained scores. Because these mechanisms are not independently varied, they remain hypotheses for future work rather than established causes.

5.6.5 Query-Level Analysis

Aggregate means can hide mixtures of repaired and introduced failures. Table 5.13 summarizes the Study A corrected seed-42 hybrid changes. The first three rows partition all 640 queries by NDCG@10 change; the final rows describe top-ten transitions and are not additional disjoint totals.

Table 5.13: Query-level changes after fine-tuning, corrected seed 42.

Outcome	Two-way hybrid	Three-way hybrid
Higher NDCG@10	102	102
Lower NDCG@10	210	163
Unchanged NDCG@10	328	375
Previously missing at ten, now retrieved	23	16
Previously retrieved at ten, now missing	48	24

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Outcome	Two-way hybrid	Three-way hybrid
Missing at ten in both systems	32	31

The two-way system repairs some top-ten misses, but introduces more than twice as many new top-ten misses as it repairs. The three-way system introduces fewer new misses, consistent with its smaller aggregate regression, but it still introduces more misses than it repairs. These counts show why the negative mean should not be interpreted as uniform deterioration, and why isolated repaired examples cannot overturn the aggregate result.

Table 5.14: Study B NDCG changes by query, using mean fine-tuned seed metric minus pretrained.

Comparison	Improved	Worse	Tied
Dense	127	96	417
Two-way	81	93	466
Three-way	85	86	469

Study B's dense-only comparison has more improved than worsened queries, while the Study B hybrids do not show the same favorable balance. This query-level view is consistent with the aggregate finding: dense improvement exists, but much of it is absorbed or neutralized when the adapted encoder is placed inside the full fusion system.

5.6.6 Overall Interpretation

The RQ3 evidence has a clear sequence. First, Study A's development selection shows large gains and justifies testing the adaptation. Second, Study A's corrected hybrid evaluation on unseen data shows regressions for both two-way and three-way hybrids. Third, Study B shows that legal fine-tuning can improve dense BGE-M3 ranking. Fourth, that dense gain largely fails to transfer into locked-weight hybrid retrieval.

This sequence supports a stronger and more nuanced conclusion than either "fine-tuning works" or "fine-tuning fails." It works at the dense-retriever level in Study B, but it does not necessarily improve the complete lexical–semantic system. The complete system must therefore be evaluated directly. Development scores and dense-only scores are insufficient substitutes for end-to-end hybrid evaluation.

Several boundaries remain. Cross-collection adaptation to ALQAC and BCA was not established. Secondary retuned-weight hybrid results were not available, so the thesis does not claim that the adapted encoder could not be made useful under a different fusion policy. A fresh untouched cohort would be required for a cleaner confirmatory test. These

limitations narrow the conclusion without weakening the observed negative transfer result.

5.7 Experimental Synthesis

The experiments answer the research questions as follows. For RQ1, hybrid retrieval improves the fixed BM25 reference in the evaluated ALQAC, Zalo, and descriptive BCA conditions. For RQ2, E5 has condition-dependent incremental value: parsed Zalo provides the clearest positive case, while ALQAC and BCA do not support a general three-way advantage. For RQ3, legal-domain adaptation improves development scores and can improve dense-only ranking, but it does not necessarily improve the complete hybrid retrieval system.

The BCA analysis adds a methodological lesson. Evaluation cannot be reduced to a ranking table when target coverage and query dependence are substantial. Missing targets, partial target availability, and giant connected components affect what a retrieval score means. Reporting both end-to-end and answerable-only views makes this limitation visible rather than hiding it inside an average.

The principal scientific takeaway is that legal retrieval systems must be evaluated at the level at which they will be used. A model can be better in isolation while adding little to a hybrid; an added retriever can help one corpus representation and not another; a retrieval failure can be caused by absent evidence rather than poor ranking. These findings justify the thesis's emphasis on controlled comparison, negative results, and careful limitation of causal claims.

Chapter 6

Conclusion and Future Work

This chapter synthesizes the experimental findings rather than repeating the numerical analysis from Chapter 5. The conclusions distinguish improvement over BM25, incremental value beyond an existing hybrid, and transfer from dense adaptation into a complete retrieval system.

6.1 Summary of Findings

6.1.1 Hybrid Retrieval over the Lexical Baseline

The experiments support the value of lexical–semantic hybrid retrieval over the fixed BM25 reference in the evaluated conditions. Both the BM25+BGE-M3 hybrid and the BM25+BGE-M3+E5 hybrid improve NDCG@10 over BM25 across the reported ALQAC and Zalo conditions, and BCA shows the same direction in its combined descriptive view. The result is strongest as an empirical finding about these collections and corpus representations. It does not imply that a single weight setting, corpus construction policy, or hybrid architecture will transfer unchanged to other legal corpora.

6.1.2 Incremental Contribution of E5

The incremental value of E5 is condition-dependent. The clearest positive result occurs on parsed Zalo, where adding E5 increases NDCG@10 by approximately 0.0130. ALQAC shows small negative differences, clean Zalo shows only a small positive difference, and BCA does not establish a reliable small-component advantage. Therefore, the thesis does not support a universal claim that three-way retrieval is better than two-way retrieval. The value of the additional dense retriever must be evaluated against a strong two-way reference in the target corpus condition.

6.1.3 Effect of Legal-Domain Adaptation

The answer is mixed and is one of the main findings of the thesis. Study A: Hybrid-Oriented Adaptation shows substantial development-selection gains, but its corrected hybrid evaluation on data unseen during training and selection shows mean NDCG@10

decreases of 0.0520 for the two-way hybrid and 0.0249 for the three-way hybrid. Study B: Dense-Oriented Adaptation shows that fine-tuning can improve BGE-M3 as a dense retriever, increasing mean NDCG@10 by 0.0189. However, that improvement largely disappears after the adapted encoder is integrated into locked-weight hybrids. Thus, dense adaptation can improve an encoder without improving the complete lexical–semantic retrieval system by a meaningful amount.

6.2 Research Contributions

The first contribution is a controlled empirical evaluation of lexical–semantic hybrid retrieval for Vietnamese legal information retrieval. The thesis compares BM25, BGE-M3, E5, two-way fusion, and three-way fusion across multiple legal collections and corpus representations. It separates improvement over a lexical baseline from incremental improvement over an already strong hybrid.

The second contribution is an evaluation of legal-domain BGE-M3 adaptation at both dense-retriever and complete-hybrid levels. The work preserves negative and non-transfer results rather than treating them as failed experiments. The distinction between dense-only improvement and hybrid-system improvement is practically important for legal retrieval systems, where the deployed system is usually a pipeline rather than a single encoder.

The third contribution is a dataset and evaluation contribution involving the BCA collection. The thesis constructs a retrieval collection from public questions and published answer citations, analyzes target coverage, and applies component-aware evaluation. This contribution shows that corpus availability and query dependence can materially affect how legal retrieval metrics should be interpreted.

6.3 Discussion and Limitations

The experiments show that hybrid retrieval is useful, but they also show why retrieval evaluation should not stop at a favorable aggregate score. A hybrid can improve over BM25 while an additional component contributes little. A model can repair some queries while introducing other failures. A benchmark can contain many questions while having fewer independent legal components than the query count suggests. These distinctions matter in legal retrieval because a system must retrieve specific provisions, not merely topically related documents.

The E5 result illustrates the limits of model-count reasoning. Adding another dense retriever increases representational diversity and expands the fusion search space, but the measured benefit depends on the collection and corpus representation. Parsed Zalo

benefits most clearly; ALQAC does not. This finding supports local validation rather than a generic architectural rule.

The adaptation results illustrate the limits of development selection. Study A's fine-tuning recipe performs well on development queries but regresses in corrected hybrid evaluation on unseen data. Study B shows that this is not simply because legal-domain fine-tuning is incapable of improving dense retrieval. Instead, the complete hybrid can absorb, dilute, or counteract dense improvements because fusion weights, lexical evidence, and score calibration all interact. The present experiments do not isolate which mechanism dominates.

The publicly available Vietnamese legal data suitable for supervised retrieval adaptation remain limited in the present setting. Additional diverse and carefully validated query–article training pairs may improve the fine-tuning process, particularly by providing broader coverage of legal topics and relevance relations. However, the study did not evaluate this intervention; it is therefore a hypothesis for future work rather than an empirical conclusion.

BCA exposes dataset limitations that are not model limitations. Some targets are absent from the indexed corpus, and some answerable queries are only partially covered. A ranking model cannot retrieve evidence that is not present. Therefore, end-to-end scores should be interpreted as measuring the whole retrieval pipeline, including corpus construction and target mapping, not only scoring quality.

Several validity limits remain. The evaluation is restricted to Vietnamese legal collections and recorded relevance judgments. It does not establish multilingual generalization, cross-jurisdiction transfer, downstream legal-answer correctness, current legal validity, user satisfaction, latency, throughput, or deployment cost. Study A's corrected adaptation evaluation on data unseen during training and selection is post-hoc because it follows a flawed preliminary analysis of that same evaluation fold. Study B also uses a fixed-corpus benchmark with prior test observation, so it should not be described as a pristine prospective confirmation.

Statistical conclusions are also bounded. Non-significant intervals do not prove equivalence, and detectable differences do not automatically imply practical importance. The thesis therefore distinguishes observed mean changes, practical improvement thresholds, and statistical uncertainty.

6.4 Future Work

A new, prospectively controlled evaluation should be conducted on an untouched query set. Corpus construction, model selection, fusion configuration, and statistical procedures should be fixed before examining the final test results.

Corpus quality should also be improved, particularly for the BCA collection. Coverage gaps can be addressed through version-aware mapping of legal documents, while query-level failures can be systematically sampled and annotated using a defined rubric. This would help distinguish missing evidence, annotation limitations, lexical mismatch, semantic confusion, and fusion-related ranking errors.

Future work should further extend the evaluation from retrieval performance to the complete RAG system. This includes assessing answer correctness, citation support, legal sufficiency, latency, throughput, and computational cost. Retrieval metrics remain useful indicators, but they cannot replace direct evaluation of generated legal answers and deployed system performance.

6.5 Closing Remarks

This thesis shows that Vietnamese legal retrieval benefits from lexical–semantic hybridization, but also that more components or better dense-only scores do not automatically produce a better complete system. The most important lesson is methodological: legal retrieval claims must be tied to the exact corpus, target coverage, selection boundary, and system level being evaluated. A disciplined negative result can be as informative as a positive result when it prevents an unsupported deployment claim.

References

[1] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge, U.K.: Cambridge University Press, 2008. [Online]. Available: https://nlp.stanford.edu/IR-book/

[2] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, "M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation," in *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 2318–2335, 2024. [Online]. Available: https://aclanthology.org/2024.findings-acl.137/

[3] L. Wang, N. Yang, X. Huang, L. Yang, R. Majumder, and F. Wei, "Multilingual E5 text embeddings: A technical report," arXiv:2402.05672, 2024. [Online]. Available: https://arxiv.org/abs/2402.05672

[4] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, 2020. [Online]. Available: https://arxiv.org/abs/2005.11401

[5] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009. [Online]. Available: https://www.nowpublishers.com/article/DownloadEBook/INR-019

[6] G. M. Rosa, R. C. Rodrigues, R. Lotufo, and R. Nogueira, "Yes, BM25 is a strong baseline for legal case retrieval," in *COLIEE 2021*, 2021. [Online]. Available: https://arxiv.org/abs/2105.05686

[7] R. Goebel, Y. Kano, M.-Y. Kim, J. Rabelo, K. Satoh, and M. Yoshioka, "Overview of benchmark datasets and methods for the Legal Information Extraction/Entailment Competition (COLIEE) 2024," in *JSAI-isAI 2024*, LNAI 14741, pp. 109–124, 2024. [Online]. Available: https://coliee.org/documents/waivers/overview_COLIEE2024.pdf

[8] ALQAC Organizers, *Automated Legal Question Answering Competition 2024*, 2024. [Online]. Available: https://sites.google.com/view/alqac-2024/home

[9] D. Q. Nguyen and A. T. Nguyen, "PhoBERT: Pre-trained language models for Vietnamese," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020. [Online]. Available: https://aclanthology.org/2020.findings-emnlp.92/

[10] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP 2019*, 2019. [Online]. Available: https://aclanthology.org/D19-1410/

[11] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. Yih, "Dense passage retrieval for open-domain question answering," in *Proc. EMNLP 2020*, pp. 6769–6781, 2020. [Online]. Available: https://aclanthology.org/2020.emnlp-main.550/

[12] O. Khattab and M. Zaharia, "ColBERT: Efficient and effective passage search via contextualized late interaction over BERT," in *Proc. 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020. [Online]. Available: https://arxiv.org/abs/2004.12832

[13] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proc. 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 758–759, 2009. [Online]. Available: https://plg.uwaterloo.ca/~gvcormac/cormacksigir09-rrf.pdf

[14] S. Bruch, S. Gai, and A. Ingber, "An analysis of fusion functions for hybrid retrieval," arXiv:2210.11934, 2022. [Online]. Available: https://arxiv.org/abs/2210.11934

[15] I. Chalkidis, M. Fergadiotis, P. Malakasiotis, N. Aletras, and I. Androutsopoulos, "LEGAL-BERT: The muppets straight out of law school," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020. [Online]. Available: https://aclanthology.org/2020.findings-emnlp.261/

[16] J. Zhan, J. Mao, Y. Liu, J. Guo, M. Zhang, and S. Ma, "Optimizing dense retrieval model training with hard negatives," in *Proc. 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 1503–1512, 2021. [Online]. Available: https://arxiv.org/abs/2104.08051

[17] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, "BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models," in *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*, 2021. [Online]. Available: https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/65b9eea6e1cc6bb9f0cd2a47751a186f-Abstract-round2.html

[18] K. Järvelin and J. Kekäläinen, "Cumulated gain-based evaluation of IR techniques," *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002. [Online]. Available: https://researchportal.tuni.fi/en/publications/cumulated-gain-based-evaluation-of-ir-techniques

[19] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," arXiv:1807.03748, 2018. [Online]. Available: https://arxiv.org/abs/1807.03748

[20] American Statistical Association, "American Statistical Association releases statement on statistical significance and p-values," 2016. [Online]. Available: https://www.amstat.org/asa/files/pdfs/p-valuestatement.pdf

[21] Government of Vietnam, "Decree No. 78/2025/ND-CP on implementation of the Law on Promulgation of Legal Normative Documents," art. 63, Apr. 1, 2025. [Online]. Available: Official Government Portal

[22] T. Chen, H. Wang, S. Chen, W. Yu, K. Ma, X. Zhao, H. Zhang, and D. Yu, "Dense X retrieval: What retrieval granularity should we use?" in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 15159–15177, 2024. [Online]. Available: https://aclanthology.org/2024.emnlp-main.845/

[23] Intfloat, "Multilingual-E5-base model card." [Online]. Available: https://huggingface.co/intfloat/multilingual-e5-base

[24] Beijing Academy of Artificial Intelligence, "BGE-M3 model card." [Online]. Available: https://huggingface.co/BAAI/bge-m3